

A Hybrid Laplacian Salp Swarm and Variable Neighborhood Search Algorithm for Continuous Wind Farm Layout Optimization

Prince Solanki, Pulkit Dwivedi, Vanita Garg, and Vinay Shukla

Abstract—The wind farm layout optimization problem (WFLOP) asks where a given number of turbines should be placed so that as little energy as possible is lost in the wakes, while every turbine stays inside the site and keeps a minimum distance from the others. In this work, we propose LX-SSA-VNS, a two-phase hybrid algorithm for the continuous WFLOP. In the first phase, the Laplacian Salp Swarm Algorithm (LX-SSA), introduced by Solanki and Deep [30], explores the layout space for half of the evaluation budget. In the second phase, the basic variable neighborhood search (VNS) refines the best layout found by the swarm by alternating shaking of the whole layout with a complete local search. Wake losses are computed with the Jensen model and direction-dependent Weibull wind data, and the boundary and spacing constraints are handled through a penalized objective. We compare LX-SSA-VNS with LX-SSA, SSA, particle swarm optimization (PSO), differential evolution (DE), VNS and a multistart SLSQP solver on 68 benchmark cases (two wind data sets, three circular farms, 2–15 turbines), using 30 seed-paired runs of 6,030 objective evaluations per method and case, and analyze the results with Friedman, Holm-corrected Wilcoxon and effect-size statistics. LX-SSA-VNS obtains the best average rank (1.89), is significantly better than each of the other six methods and finds a feasible layout in 99.9% of the runs. An ablation with SSA, LX-SSA, VNS and SSA-VNS shows that the improvement comes from pairing a swarm phase with VNS, which removes on average 39% of the wake loss left by the swarm, whereas the Laplace step itself adds nothing: an SSA first phase does at least as well. On the Horns Rev 1 offshore site, the hybrid gives the highest mean energy yield for a 16-turbine block, although no method improves on the installed layout within the budget and only the constraint-aware SLSQP solver finds feasible layouts for the full 80-turbine farm. Finally, we show that the ranking of the algorithms is stable under a different power curve but changes under a Gaussian wake model.

Index Terms—Hybrid metaheuristic, Laplacian Salp Swarm Algorithm (LX-SSA), Variable neighborhood search, Wake effect, Wind farm layout optimization.

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I. INTRODUCTION

Wind energy has become one of the main sources of low-carbon electricity. Wind turbines convert the kinetic energy of moving air into electrical energy without direct greenhouse gas emissions, and the installed capacity has grown quickly over the last two decades. Since most of the cost of a wind project is fixed at the planning stage, the decisions taken before construction have a lasting effect on the energy a farm will deliver, and one of the most important of these decisions is where the turbines are placed.

The Wind Farm Layout Optimization Problem (WFLOP) deals with exactly this question. An upstream turbine extracts momentum from the flow and leaves behind a wake of slower and more turbulent air, so that the turbines downstream produce less power. The wake widens and recovers with distance, which suggests spreading the turbines as far apart as possible. The site, however, is limited, and the turbines must keep a minimum distance from each other. Placing more turbines in the same area raises the installed capacity but also strengthens the wake interaction, so that the energy produced per turbine falls. The objective is nonlinear, the wakes couple the positions of all turbines, and the feasible region is bounded by many nonconvex spacing constraints. For these reasons, WFLOP has become a popular test bed for metaheuristic search.

Mosetti et al. [1] and Grady et al. [2] were among the first to address WFLOP, using a binary-coded genetic algorithm (GA) on a grid to maximize the energy output at minimum installation cost. Ozturk and Norman [3] extended this work with a different objective and a heuristic that places turbines in continuous space, taking wind speed, wind direction and wake interactions into account. Lackner and Elkinton [4] proposed an analytical model for offshore farms in which the annual power output depends directly on the turbine positions and wakes. Castro et al. [5] designed layouts with a GA under spacing and wind-speed constraints, and Huang [6] used a distributed GA to maximize the annual profit of large farms from their energy output, capital cost and operating cost. Elkinton et al. [7] compared GA, particle swarm optimization (PSO), differential evolution (DE) and simulated annealing (SA) for offshore layouts. SA was also used by Bilbao and Alba [8] to maximize the annual profit and by Rivas et al. [9], whose version combines three local-search moves for large offshore farms.

Emami and Noghereh [10] added the construction cost to the objective and solved the resulting problem with a GA, while

Sisbot et al. [11] used a multi-objective GA to trade energy production against installation cost. Kusiak and Song [12] introduced the wind-distribution-based placement model that we also use in this paper. They treated the wake loss as a function of the turbine locations and the wind direction and solved a bi-criteria version of the problem with a multi-objective evolution strategy. Eroglu et al. [13], [14] applied ant colony optimization (ACO) and particle filtering to the same model, Samorani [15] studied the trade-off between power output and wake effect, and Bansal and Farswan [16] used biogeography-based optimization (BBO) and proposed an upper limit on the number of turbines for a given farm. ACO has also been used to design the collector cable system of offshore farms [27].

Later studies refined both the algorithms and the models. Ju and Liu [17] proposed adaptive and self-informed variants of the GA, and Jin et al. [18] combined a wake-aware power-loss cost model with an adaptive PSO. Hou et al. [28] reviewed the layout and electrical-system design of wind farms, including the routing of cables. Dhiman et al. [19] used lidar simulations to locate the wakes and steer the yaw of downstream turbines, Nagpal et al. [20] refined the layouts of a distributed GA with a continuous local method, and PSO has been applied to reduce wake losses and increase the power output [29]. Bai et al. [21] integrated Monte Carlo tree search into an adaptive GA and applied it to a planned wind farm in New Jersey, and Cazzaro and Pisinger [22] developed a variable neighborhood search (VNS) heuristic that handles spacing constraints and construction costs. Beyond classical metaheuristics, pseudo-gradient methods have been developed for large design spaces [33], three-dimensional Gaussian wake models have been coupled with layout optimization [34], and machine-learning [35], CFD-based Kriging [36] and data-driven or physics-informed [37], [38] surrogates have been introduced to reduce the cost of higher-fidelity evaluations. No single optimizer or wake model dominates this literature; each study balances model fidelity, computing cost and scalability in its own way.

Swarm methods such as the Salp Swarm Algorithm (SSA) [26] are attractive for WFLOP because they are simple and cheap per iteration. Solanki and Deep [30] introduced the Laplacian Salp Swarm Algorithm (LX-SSA), in which each follower salp also tries a Laplace-distributed move relative to the food source and keeps the better of its two positions. The heavy tail of the Laplace distribution produces mostly short moves and occasionally long ones, which helps the swarm to visit separated regions of the layout space. In WFLOP, however, we observed that a swarm spends most of the second half of its budget on small improvements once it has contracted around the best layout, because the final refinement of a layout requires coordinated moves of single turbines along the spacing constraints. Trajectory methods such as VNS [40], [41] behave the other way round: they refine one layout very efficiently by alternating random perturbations of growing size with a local search, but the result depends on where the search starts.

This complementary behavior motivates the present work. We propose LX-SSA-VNS, a two-phase hybrid in which

LX-SSA explores the layout space for the first half of the evaluation budget and the basic VNS then refines the best layout found by the swarm. Both phases work on the same continuous turbine-coordinate representation and the same penalized objective. We keep the standard ingredients of the WFLOP benchmark, namely the Jensen wake model, a Weibull description of the wind and the Kusiak–Song power model, so that our results can be compared with the earlier literature, and we examine the effect of these modeling choices separately in Section VIII. Our contributions are summarized as follows:

- 1) We propose LX-SSA-VNS, a two-phase hybrid in which the previously published LX-SSA [30] explores the layout space and the basic VNS [40], [42] refines the best layout (Section V).
- 2) We formulate the continuous WFLOP with explicit boundary and minimum-spacing constraints, handled through a penalized objective that both phases of the hybrid share (Section II).
- 3) We compare the hybrid with LX-SSA, SSA, PSO, DE, VNS and a gradient-based multistart SLSQP solver on 68 benchmark cases at equal numbers of objective evaluations, with 30 seed-paired runs per method and case, and support the comparison with Friedman, Holm-corrected Wilcoxon and effect-size statistics, feasibility rates and convergence curves (Section VII).
- 4) We carry out a component ablation with SSA, LX-SSA, VNS, SSA-VNS and LX-SSA-VNS that separates the effects of the swarm phase, the Laplace step and the VNS phase, and we examine how the budget should be split between the two phases (Section VII-F).
- 5) We apply all methods to the Horns Rev 1 offshore farm with its real turbine, wind climate and site outline (Section VII-I), and we test how the results depend on the power curve, the wake model, the minimum spacing and the boundary handling (Section VIII).

The remainder of this paper is organized as follows. Section II describes the WFLOP model and the constraint handling. Section III summarizes LX-SSA, and Section IV explains how it is applied to WFLOP. Section V introduces the hybrid LX-SSA-VNS. Section VI describes the experimental setup, and Section VII reports the numerical results, including the ablation and the Horns Rev 1 case. Section VIII examines the sensitivity of the results to the modeling assumptions, Section IX discusses the computational complexity, and Section X states the limitations of the study. Finally, Section XI concludes the paper.

II. PROBLEM MODELING

We use the following definitions and assumptions for the design of a wind farm.

- 1) **Prescribed turbine count:** The nameplate capacity of a farm of identical turbines grows linearly with the number of turbines, but the expected farm power does not, because the wake losses depend on the layout. In every optimization run the number of turbines N is fixed in advance, and only the $2N$ turbine coordinates are optimized. Results for several values of N therefore

form a sweep over the turbine count; N and the turbine positions are not optimized jointly.

- 2) **Turbine location:** Each turbine is located by its planar coordinates (ρ, σ) , measured from the center of the farm.
- 3) **Identical turbines:** All turbines have the same power curve, rated capacity, rotor size and hub height.
- 4) **Wind speed:** For a given site, height and wind direction, the wind speed s follows a Weibull distribution with probability density function

$$p_s(s; \zeta, \psi) = \frac{\zeta}{\psi} \left(\frac{s}{\psi} \right)^{\zeta-1} \exp \left(- \left(\frac{s}{\psi} \right)^\zeta \right),$$

where ζ is the shape parameter and ψ the scale parameter.

- 5) **Wind direction:** The wind speed and both Weibull parameters depend on the wind direction θ , i.e., $s = s(\theta)$, $\zeta = \zeta(\theta)$ and $\psi = \psi(\theta)$ with $0^\circ \leq \theta \leq 360^\circ$. Fig. 1 shows the convention used throughout the paper: $\theta = 0^\circ$ points east, 90° north, 180° west and 270° south.
- 6) **Minimum spacing:** Any two turbines must be at least $4D = 8R$ apart, where $D = 2R$ is the rotor diameter:

$$(\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2 \geq 64R^2.$$

We use this value because it is the one adopted in the benchmark studies we compare with [12], [16]. Real projects often require larger and direction-dependent distances, and the spacing limits how many turbines fit into a farm; Section VIII-B studies the effect of $5D$ and $6D$ spacing.

- 7) **Farm shape:** The farm is circular, as in the benchmark studies [12], [16]. A circle has no preferred orientation, so the boundary itself favors no wind direction. We make no claim that a circular site uses land better than other shapes.
- 8) **Farm boundary:** Every turbine $T_i(\rho_i, \sigma_i)$ must lie inside the farm of radius r ,

$$\rho_i^2 + \sigma_i^2 \leq r^2.$$

We study farms with $r = 500, 750$ and 1000 m.

- 9) **Search space:** The search space is the circular farm area shown in Fig. 1. Because it is continuous, the turbines are not restricted to the nodes of a grid.
- 10) **Model:** The WFLOP model consists of a wake model, which describes the speed reduction behind each turbine, and a power model. We use the Jensen wake model [23], [24] and the power model of Kusiak and Song [12].
- 11) **Objective:** The aim is to maximize the expected power of the farm, i.e., to minimize the wake losses, subject to the spacing and boundary constraints of assumptions 6 and 8.

A. Wake Effect Model

A turbine extracts momentum from the wind, so the air behind it moves more slowly and the downstream turbines produce less power. Accounting for these wake losses is therefore at the core of layout design. We use Jensen's wake model [23], [24], which is cheap to evaluate and has been used

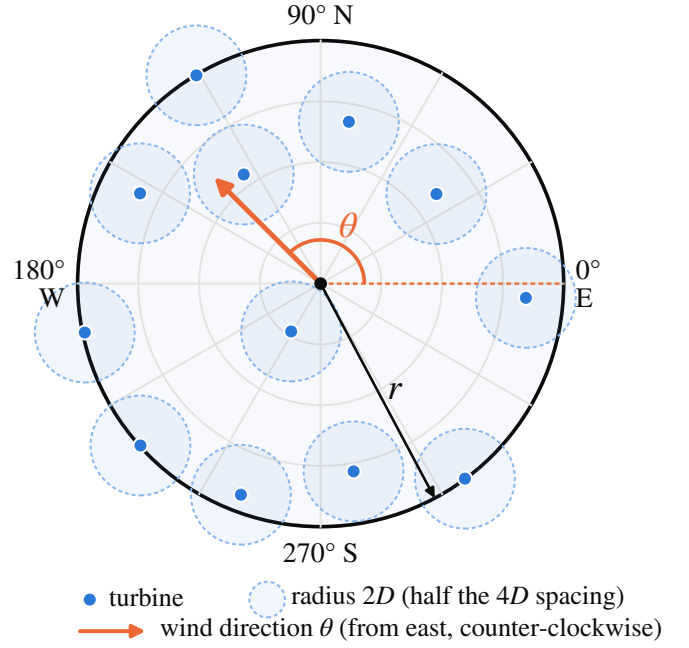


Fig. 1: Circular wind farm of radius r . The wind direction θ is measured counter-clockwise from east. The points show the best LX-SSA-VNS layout for 12 turbines in the 750-m farm (Wind Data Set I); the shaded discs have radius $2D$, so two discs touch when their turbines are exactly $4D$ apart.

in most of the benchmark studies we compare with. Gaussian and three-dimensional wake models describe the velocity field more realistically and have also been used for layout optimization [34], [39]; for this reason, Section VIII-A re-evaluates all optimized layouts with a Gaussian wake model.

In the Jensen model, the wake behind a rotor expands linearly with the downstream distance, and the wind speed inside it drops from the upstream value s_{up} to s_{down} (Fig. 2). Let K be the wake spreading constant, R the rotor radius and d_{ij} the distance between turbines i and j along the wind direction. The velocity deficit induced at turbine j by the wake of turbine i is

$$\delta s_{ij} = 1 - (s_{\text{down}}/s_{\text{up}}) = \frac{1 - \sqrt{1 - C_T}}{\left(1 + \frac{K d_{ij}}{R}\right)^2},$$

where C_T is the thrust coefficient of the turbine.

When a turbine lies in the wakes of several upstream turbines, the individual deficits are combined by the root-sum-square rule. For a turbine i that is affected by the wakes of the other turbines,

$$\delta s_i = \sqrt{\sum_{\substack{j=1 \\ j \neq i}}^N (\delta s_{ij})^2}. \quad (1)$$

Each rotor faces the incoming wind direction θ . The wake of a turbine at (ρ, σ) can then be idealized as a half cone whose virtual vertex A lies a distance R/K upstream of the rotor (Fig. 3). The half-cone angle α ($0 \leq \alpha \leq \pi/2$), i.e., the angle between the axis and the side of the cone, is $\alpha = \arctan(K)$.

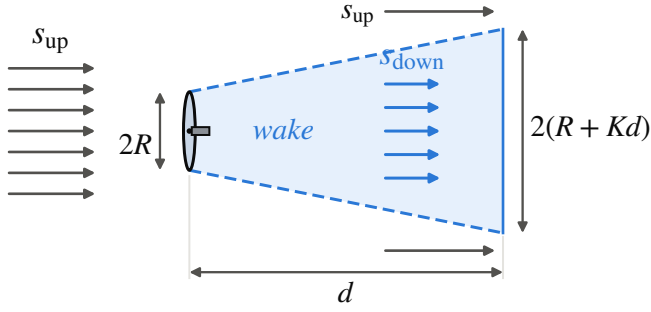


Fig. 2: Jensen wake behind a rotor of radius R (schematic, not to scale). The wake widens linearly with the spreading constant K , so that its width at the downstream distance d is $2(R + Kd)$, and the wind speed inside it drops from s_{up} to s_{down} .

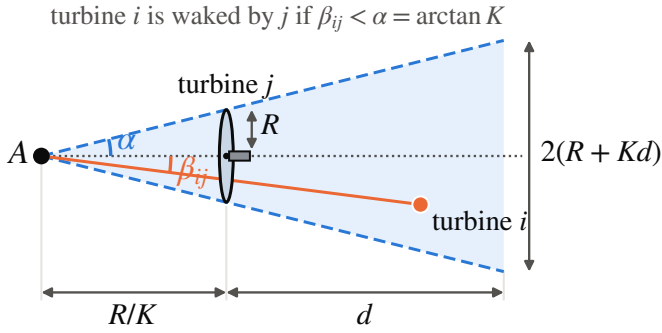


Fig. 3: Wake half cone of turbine j (schematic, K exaggerated). The virtual vertex A lies a distance R/K upstream of the rotor, the half-cone angle is $\alpha = \arctan K$, and a downstream turbine i lies in the wake of j when $\beta_{ij} < \alpha$.

A turbine lies in the wake of another if it falls inside this cone, which the following two lemmas make precise.

Lemma 1. For a given wind direction θ , let β_{ij} ($0 \leq \beta_{ij} \leq \pi$) be the angle at the vertex A of the wake of turbine j between the wake axis and the line from A to turbine i . Then

$$\beta_{ij} = \cos^{-1} \left(\frac{u_{ij}}{\sqrt{u_{ij}^2 + v_{ij}^2}} \right),$$

with

$$\begin{aligned} u_{ij} &= (\rho_i - \rho_j) \cos \theta + (\sigma_i - \sigma_j) \sin \theta + \frac{R}{K}, \\ v_{ij} &= -(\rho_i - \rho_j) \sin \theta + (\sigma_i - \sigma_j) \cos \theta, \end{aligned}$$

where R/K is the distance from the rotor center to the vertex A .

Lemma 2. If turbine i lies in the wake of turbine j , the distance between the two along the wind direction θ is

$$d_{ij} = |(\rho_i - \rho_j) \cos \theta + (\sigma_i - \sigma_j) \sin \theta|.$$

Restricting the sum in (1) to the turbines j whose wake contains turbine i , i.e., to those with $\beta_{ij} < \alpha$, gives

$$\delta s_i = \sqrt{\sum_{\substack{j=1 \\ j \neq i, \beta_{ij} < \alpha}}^N (\delta s_{ij})^2}. \quad (2)$$

Equation (2) shows that δs_i depends on the wind direction θ and on the position (ρ_i, σ_i) of turbine i .

The wake deficit enters the energy calculation through the Weibull scale parameter. For a direction bin θ , let the free-stream speed be

$$S_\theta \sim \text{Weibull}(\zeta(\theta), \psi(\theta)).$$

In the Jensen benchmark, the combined deficit $\delta s_i(\theta)$ acts as a fixed multiplicative reduction within the bin, so the local inflow at turbine i is

$$S_{i,\theta} = [1 - \delta s_i(\theta)] S_\theta.$$

Since $aS \sim \text{Weibull}(\zeta, a\psi)$ for any constant $a > 0$ when $S \sim \text{Weibull}(\zeta, \psi)$, the shape parameter is unchanged and the scale parameter becomes

$$\psi_i(\theta) = \psi(\theta) [1 - \delta s_i(\theta)], \quad i = 1, 2, \dots, N. \quad (3)$$

This relation holds under the assumption that the wake acts as a deterministic speed multiplier in each direction bin. More detailed wake and turbulence models would change the whole speed distribution and not only its scale.

B. Power Model

We use the simplified power curve of the Kusiak–Song benchmark [12],

$$f(s) = \begin{cases} 0, & s < s_{\text{cut-in}}, \\ \lambda' s + \eta, & s_{\text{cut-in}} \leq s \leq s_{\text{rated}}, \\ P_{\text{rated}}, & s > s_{\text{rated}}. \end{cases} \quad (4)$$

where $s_{\text{cut-in}}$ and s_{rated} are the cut-in and rated speeds, and λ' and η define the linear part of the curve. The benchmark does not include a cut-out speed, and we use the model exactly as written. A linear curve is only a rough approximation of a real turbine, so in Section VIII-A we re-evaluate all layouts with a cubic power curve, with and without a 25 m/s cut-out.

For a turbine at (ρ, σ) and wind direction θ , the expected power output is

$$\begin{aligned} E(P_i) &= \int_0^{360} P_\theta(\theta) \cdot E(P_i, \theta) d\theta \\ &= \int_0^{360} P_\theta(\theta) \cdot \int_0^\infty f(s) \cdot p_s(s, \zeta(\theta), \psi_i(\theta)) ds d\theta. \end{aligned} \quad (5)$$

The layout problem is then to maximize the total expected power of the farm subject to the spacing and boundary constraints of assumptions 6 and 8:

$$\begin{aligned} &\text{Maximize} \quad \sum_{i=1}^N E(P_i) \\ &\text{subject to} \quad (\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2 \geq 64R^2, \\ &\quad \quad \quad \rho_i^2 + \sigma_i^2 \leq r^2, \quad i, j = 1, 2, \dots, N, \quad i \neq j \end{aligned}$$

where $E(P_i)$ is the expected power output of the i th turbine. Substituting (4) and the Weibull density into (5) gives

$$\begin{aligned} E(P_i) &= \int_0^{360} p_\theta(\theta) \int_0^\infty f(s) p_s(s, \zeta(\theta), \psi_i(\theta)) ds d\theta \\ &= \int_0^{360} p_\theta(\theta) \int_0^\infty \frac{f(s) \zeta(\theta)}{\psi_i(\theta)} \left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)-1} e^{-\left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)}} ds d\theta \\ &= \int_0^{360} p_\theta(\theta) \left[\lambda' \int_{s_{\text{cut-in}}}^{s_{\text{rated}}} \frac{s \zeta(\theta)}{\psi_i(\theta)} \left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)-1} e^{-\left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)}} ds \right. \\ &\quad \left. + \eta \int_{s_{\text{cut-in}}}^{s_{\text{rated}}} \frac{\zeta(\theta)}{\psi_i(\theta)} \left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)-1} e^{-\left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)}} ds \right. \\ &\quad \left. + P_{\text{rated}} \int_{s_{\text{rated}}}^\infty \frac{\zeta(\theta)}{\psi_i(\theta)} \left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)-1} e^{-\left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)}} ds \right]. \end{aligned}$$

Following Kusiak and Song [12], we evaluate these integrals with a Riemann sum [25]. The wind direction is divided into bins $\theta_0 = 0^\circ, \theta_1, \dots, \theta_{N_\theta+1} = 360^\circ$, and the speed range between $s_{\text{cut-in}}$ and s_{rated} into bins $s_0 = s_{\text{cut-in}}, s_1, \dots, s_{N_s+1} = s_{\text{rated}}$, all of equal width. The expected power of the i th turbine is then approximated by

$$\begin{aligned} E(P_i) &= \lambda' \sum_{j=1}^{N_s+1} \left(\frac{s_{j-1} + s_j}{2} \right) \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} \\ &\quad \left[e^{-\left(\frac{s_{j-1}}{\psi_i \left(\frac{\theta_l + \theta_{l-1}}{2} \right)} \right)^{\zeta \left(\frac{\theta_l + \theta_{l-1}}{2} \right)}} - e^{-\left(\frac{s_j}{\psi_i \left(\frac{\theta_l + \theta_{l-1}}{2} \right)} \right)^{\zeta \left(\frac{\theta_l + \theta_{l-1}}{2} \right)}} \right] \\ &\quad + P_{\text{rated}} \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} e^{-\left(\frac{s_{\text{rated}}}{\psi_i \left(\frac{\theta_l + \theta_{l-1}}{2} \right)} \right)^{\zeta \left(\frac{\theta_l + \theta_{l-1}}{2} \right)}} \\ &\quad + \eta \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} \\ &\quad \left[e^{-\left(\frac{s_{\text{cut-in}}}{\psi_i \left(\frac{\theta_l + \theta_{l-1}}{2} \right)} \right)^{\zeta \left(\frac{\theta_l + \theta_{l-1}}{2} \right)}} - e^{-\left(\frac{s_{\text{rated}}}{\psi_i \left(\frac{\theta_l + \theta_{l-1}}{2} \right)} \right)^{\zeta \left(\frac{\theta_l + \theta_{l-1}}{2} \right)}} \right]. \end{aligned}$$

where N_s and N_θ are the numbers of speed and direction intervals and ω_{l-1} is the probability that the wind blows from the $(l-1)$ th direction bin.

a) Benchmark objective and energy: Like the earlier studies, we keep the bin-width factor $(\theta_l - \theta_{l-1})$ in the sum. The objective values in our tables are therefore benchmark values and not electrical power bounded by NP_{rated} . Because all 24 direction bins are 15° wide and $\sum_l \omega_{l-1} = 1$, the objective equals 15 times the expected farm power in kW, and the annual energy production follows as AEP [MWh] = (objective/15) $\times$ 8.76. For example, the wake-free objective of a single turbine under Wind Data Set I is 14045.74, which corresponds to an expected power of 936.38 kW, a capacity factor of 0.624 and an AEP of 8202.7 MWh. We confirmed this conversion with the independent evaluator described in Section VI-E.

C. Constraint Handling

Every layout must satisfy the circular boundary and the minimum spacing. In all our runs, both constraints are handled by two simple operations, a box repair of the coordinates and

a large additive penalty in the minimized objective; no other repair operator is used.

1) Box Repair: After every position update, each coordinate is clipped to the square that encloses the farm,

$$\begin{aligned} \rho_i &\leftarrow \min\{\max\{\rho_i, -r\}, r\}, \\ \sigma_i &\leftarrow \min\{\max\{\sigma_i, -r\}, r\}. \end{aligned}$$

This keeps the decision variables bounded, but it does not guarantee $\rho_i^2 + \sigma_i^2 \leq r^2$; the circular boundary is enforced by the penalty below.

2) Constraint Violations: The spacing and boundary constraints are $d_{ij} = \sqrt{(\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2} \geq d_{\min} = 8R$ for all $i \neq j$ and $\rho_i^2 + \sigma_i^2 \leq r^2$ for all i . We measure their violations by

$$g_{ij}^s = d_{\min} - d_{ij}, \quad g_i^b = \rho_i^2 + \sigma_i^2 - r^2, \quad (6)$$

and call a layout feasible if $g_{ij}^s \leq 0$ and $g_i^b \leq 0$ for all i and j .

3) Penalized Objective: All penalty-based optimizers minimize the penalized wake loss

$$\begin{aligned} F_p &= \left(NE_{\text{ideal}} - \sum_{i=1}^N E(P_i) \right) + \sum_{i: g_i^b > 0} (1 + \lambda g_i^b)^2 \\ &\quad + \sum_{i < j: g_{ij}^s > 0} (1 + \lambda g_{ij}^s)^2, \quad \lambda = 10^{10}, \end{aligned} \quad (7)$$

where NE_{ideal} is the wake-free objective, so that the first term is the wake loss reported in our tables. Any violation larger than about 10^{-7} (m or m^2) makes the penalty larger than the largest possible wake loss. Comparing F_p values therefore gives a feasibility-first selection: a feasible layout is always preferred to an infeasible one, two feasible layouts are compared by their wake loss, and two infeasible layouts by their total penalty. Box repair and F_p are used in every update, comparison, sort and greedy selection of Algorithms 1 and 2. Section VIII-C shows that a radial projection onto the circle, used instead of box clipping, changes the results noticeably, so we keep the same boundary rule for all methods.

III. PRELIMINARIES: LAPLACIAN SALP SWARM ALGORITHM

LX-SSA was introduced by Solanki and Deep [30], and we summarize here only the operators needed to follow and reproduce its use for WFLOP. As in SSA [26], the sorted population is split into a leader half and a follower half, and the best solution found so far is kept as the food source H . LX-SSA keeps these SSA dynamics and adds a Laplace-based candidate with greedy selection to the follower update [30], [31].

A leader salp $i \leq N_p/2$ updates its j th coordinate around the food source as

$$\rho_j^i = \begin{cases} H_j + r_1((ub_j - lb_j)r_2 + lb_j), & r_3 \geq 0.5, \\ H_j - r_1((ub_j - lb_j)r_2 + lb_j), & r_3 < 0.5, \end{cases} \quad (8)$$

where $r_2, r_3 \sim U(0, 1)$ and

$$r_1 = 2 \exp \left[- \left(\frac{4l}{L} \right)^2 \right],$$

with iteration l and maximum iteration L . A follower salp $i > N_p/2$ first forms the standard SSA position from itself and the preceding salp,

$$\rho_j^i = \frac{1}{2} (\rho_j^i + \rho_j^{i-1}). \quad (9)$$

In addition, LX-SSA forms a second follower candidate

$$y_j^i = x_j^i + \gamma(z, \phi, \chi) (H_j - x_j^i), \quad (10)$$

where x_j^i is the current coordinate of the follower and

$$\gamma(z, \phi, \chi) = \begin{cases} \phi + \chi \ln(2z), & z \leq 0.5, \\ \phi - \chi \ln(2(1-z)), & z > 0.5, \end{cases} \quad (11)$$

with $z \sim U(0, 1)$. This is the inverse-distribution sample of a Laplace variable with location ϕ and scale χ (mean ϕ , variance $2\chi^2$). The location ϕ centers the step, and the scale $\chi > 0$ controls its spread: a small χ keeps the moves short, while a large χ makes long exploratory moves more likely. We use $\phi = 0$ and $\chi = 1$ in all experiments, so that γ is symmetric about zero and $P(|\gamma| > 1) = e^{-1} \approx 0.37$. Most candidates therefore move towards H ($\gamma > 0$) or away from it ($\gamma < 0$) by a fraction of their current distance, while the heavy tail occasionally produces a move beyond H or far away from it. These values were not tuned for WFLOP. The follower keeps the better of its two candidates. Since each follower evaluates both candidates before the population is re-evaluated, one LX-SSA iteration costs $N_p/2 \times 2 + N_p = 2N_p$ objective calls, twice the cost of an SSA iteration, and all comparisons in Section VII are therefore made at equal numbers of calls.

Algorithm 1 summarizes the search. Whether one candidate is better than another is decided by the objective and, for WFLOP, by the feasibility rules of Section II-C. Because SSA and LX-SSA differ only in the Laplace-based follower candidate, a comparison between the two isolates the effect of this mechanism.

Algorithm 1 Previously developed LX-SSA search logic [30]

- 1: Initialize and evaluate the salp population; sort it and set the best solution as food source H .
 - 2: **while** the termination criterion is not met **do**
 - 3: Update r_1 .
 - 4: **for** salp $i = 1, \dots, N_p$ **do**
 - 5: **if** $i \leq N_p/2$ **then**
 - 6: Update leader i using (8).
 - 7: **else**
 - 8: Form the standard follower candidate using (9).
 - 9: Draw $\gamma(z, \phi, \chi)$ using (11).
 - 10: Form the Laplace-based follower candidate using (10).
 - 11: Evaluate both follower candidates and retain the better one.
 - 12: **end if**
 - 13: **end for**
 - 14: Enforce variable bounds, re-evaluate the population, sort, and update H .
 - 15: **end while**
 - 16: **return** the best solution H .
-

IV. APPLICATION OF LX-SSA TO WFLOP

We apply the published LX-SSA [30] to WFLOP without changing the algorithm. For a prescribed number of turbines N , candidate j is the vector of all turbine coordinates,

$$\mathbf{x}_j = (\rho_j^1, \sigma_j^1, \rho_j^2, \sigma_j^2, \dots, \rho_j^N, \sigma_j^N) \in \mathbb{R}^{2N},$$

so the optimizer searches only the turbine positions, and N stays fixed during a run.

Algorithm 2 lists the steps of the implementation. The candidate coordinates are updated first and box-repaired; the wake deficits and Weibull scale factors are then recomputed for every direction bin, the constraint violations are measured, and the penalized objective F_p is evaluated. All comparisons use F_p , which gives the feasibility-first selection of Section II-C. The order of these steps matters because the energy of each turbine depends on the wake field of the whole layout.

Algorithm 2 Application of LX-SSA to WFLOP for a prescribed turbine count N

- Require:** Farm radius r , prescribed N , wind data, turbine/wake parameters, population size N_p , maximum iterations T , ϕ , χ
- 1: Set dimension $d = 2N$, bounds $[-r, r]^d$, and draw N_p coordinate vectors uniformly in the bounds.
 - 2: **for** each candidate **do**
 - 3: For each wind-direction bin, compute wake deficits $\delta s_i(\theta)$.
 - 4: Set $\psi_i(\theta)$ using (3) and compute $E(P_i)$.
 - 5: Compute the violations (6) and the penalized objective F_p from (7).
 - 6: **end for**
 - 7: Sort the population by F_p and set the best candidate as food source H .
 - 8: **for** $t = 1$ to T **do**
 - 9: Generate the leader and follower candidates of Algorithm 1, applying box repair to each; followers keep the better of their two candidates by F_p .
 - 10: Apply box repair, evaluate F_p for the new population and sort it.
 - 11: If the best new candidate has a smaller F_p than H , replace H .
 - 12: **end for**
 - 13: **return** H and its turbine coordinates, objective value, wake loss and feasibility status.
-

V. HYBRID LX-SSA-VNS ALGORITHM

Swarm methods such as LX-SSA spread their search over many regions of the layout space, but once the swarm has contracted around the food source, most of the remaining evaluations bring only small gains. VNS [40], [41] behaves the other way round: it refines a single layout systematically, alternating random perturbations of increasing size with a complete local search, but its result depends on the starting point. We combine the two in a two-phase hybrid, LX-SSA-VNS, in which LX-SSA [30] explores the layout space and the basic VNS refines the best layout it finds.

TABLE I: Wind Data Sets I and II: Direction Intervals, Weibull Parameters, and Blowing Probabilities

Int.	Start (°)	End (°)	ζ	ψ	Prob.
1	0	15	2	13	0.000
2	15	30	2	13	0.010
3	30	45	2	13	0.010
4	45	60	2	13	0.010
5	60	75	2	13	0.010
6	75	90	2	13	0.200
7	90	105	2	13	0.600
8	105	120	2	13	0.010
9	120	135	2	13	0.010
10	135	150	2	13	0.010
11	150	165	2	13	0.010
12	165	180	2	13	0.010
13	180	195	2	13	0.010
14	195	210	2	13	0.010
15	210	225	2	13	0.010
16	225	240	2	13	0.010
17	240	255	2	13	0.010
18	255	270	2	13	0.010
19	270	285	2	13	0.010
20	285	300	2	13	0.010
21	300	315	2	13	0.010
22	315	330	2	13	0.010
23	330	345	2	13	0.010
24	345	360	2	13	0.000

Int.	Start (°)	End (°)	ζ	ψ	Prob.
1	0	15	2	7.0	0.0002
2	15	30	2	5.0	0.0080
3	30	45	2	5.0	0.0227
4	45	60	2	5.0	0.0242
5	60	75	2	5.0	0.0225
6	75	90	2	4.0	0.0339
7	90	105	2	5.0	0.0423
8	105	120	2	6.0	0.0290
9	120	135	2	7.0	0.0617
10	135	150	2	7.0	0.0813
11	150	165	2	8.0	0.0994
12	165	180	2	9.5	0.1394
13	180	195	2	10.0	0.1839
14	195	210	2	8.5	0.1115
15	210	225	2	8.5	0.0765
16	225	240	2	6.5	0.0080
17	240	255	2	4.6	0.0051
18	255	270	2	2.6	0.0019
19	270	285	2	8.0	0.0012
20	285	300	2	5.0	0.0010
21	300	315	2	6.4	0.0017
22	315	330	2	5.2	0.0031
23	330	345	2	4.5	0.0097
24	345	360	2	3.9	0.0317

A. Phase 1: Exploration by LX-SSA

For a budget of B objective calls and a split fraction $\rho \in (0, 1)$, LX-SSA (Algorithm 2) runs with population N_p for

$$T_1 = \text{round}\left(\frac{\rho B - N_p}{2N_p}\right)$$

iterations, since one LX-SSA iteration costs $2N_p$ calls. Its parameter r_1 decreases over these T_1 iterations, so the swarm itself moves from exploration to exploitation within Phase 1. The food source H at the end of Phase 1 becomes the starting point of Phase 2.

B. Phase 2: Refinement by Basic VNS

Phase 2 is the basic VNS of Mladenović and Hansen [40], [41] in the continuous form of Mladenović et al. [42], applied to the penalized objective F_p in (7). It uses $k_{\max} = 5$ nested ℓ_∞ neighborhoods of the whole layout,

$$\mathcal{N}_k(x) = \{y : \rho_{k-1} < \|y - x\|_\infty \leq \rho_k\}, \quad \rho_k = 0.1 k r.$$

In the shaking step, a point is drawn uniformly from $\mathcal{N}_k(x)$, so that all $2N$ coordinates move, and box repair is applied. The shaken layout is then improved by a complete best-improvement local search. Since the Jensen top-hat wake makes F_p discontinuous, we use a derivative-free compass search: it evaluates all $4N$ moves $\pm h$ along the coordinate axes, moves to the best improving one, and halves h (starting from $0.05r$) when no move improves, until $h < 10^{-3}r$. If the resulting local minimum is better than the incumbent, it is accepted and k is reset to 1; otherwise k is increased by one, and it returns to 1 after $k_{\max}$.

Phase 2 first applies the local search to H and then repeats shaking, local search and neighborhood change until the budget B is used up. It returns the best layout evaluated. Box

repair and the penalized objective are used exactly as in LX-SSA, so the feasibility-first rules of Section II-C hold in both phases. Algorithm 3 gives the complete procedure.

C. Parameters and Cost

The hybrid uses $N_p = 30$, $\phi = 0$, $\chi = 1$ and the VNS settings above, and it adds a single parameter, the split ρ . With the default $\rho = 0.5$, LX-SSA receives 3,030 of the 6,030 calls ($T_1 = 50$) and VNS the remaining 3,000; the effect of ρ is studied in Section VII-F. The cost of both phases is dominated by the objective evaluations, so for a budget of B calls the hybrid has the same $\mathcal{O}(B N^2 N_\theta N_s)$ cost as the other methods. Because the random-number generator is seeded once and Phase 1 draws the initial population first, the hybrid starts from the same initial population as all other methods for a given seed, which keeps our comparisons paired.

VI. EXPERIMENTAL SETUP

A. Test Problems

We use two wind data sets (Table I). In Wind Data Set I, the Weibull parameters are the same in all directions ($\zeta = 2$, $\psi = 13$) and the wind is strongly directional: $\omega_0 = \omega_{23} = 0$, $\omega_5 = 0.2$, $\omega_6 = 0.6$, and $\omega_l = 0.01$ for all other intervals. Wind Data Set II keeps $\zeta = 2$, but the scale parameter varies with direction, from $\psi = 2.6$ (255–270°) to $\psi = 10$ (180–195°), and the wind is spread over more directions. The wind direction is divided into $N_\theta = 24$ bins of 15°, and the speed range from $s_{\text{cut-in}}$ to s_{rated} into $N_s = 21$ intervals of 0.5 m/s.

Each data set is combined with three circular farms of radius $r = 500, 750$ and 1000 m, and the turbine count is prescribed for each run: $N = 2, \dots, 10$ for 500 m, $N = 2, \dots, 12$ for 750 m and $N = 2, \dots, 15$ for 1000 m. This gives $2 \times (9 + 11 + 14) = 68$ test cases. The turbine and wake parameters are:

Algorithm 3 Hybrid LX-SSA-VNS for a prescribed turbine count N

Require: budget B , split ρ , population N_p , LX-SSA parameters ϕ, χ , farm radius r

- 1: $T_1 \leftarrow \text{round}((\rho B - N_p)/(2N_p))$
- 2: Run LX-SSA (Algorithm 2) for T_1 iterations; let H be its food source
- 3: $x \leftarrow \text{LOCALSEARCH}(H)$; $k \leftarrow 1$
- 4: **while** fewer than B objective calls have been used **do**
- 5: $y \leftarrow$ uniform random point of $\mathcal{N}_k(x)$; box repair
- 6: $y \leftarrow \text{LOCALSEARCH}(y)$
- 7: **if** $F_p(y) < F_p(x)$ **then**
- 8: $x \leftarrow y$; $k \leftarrow 1$
- 9: **else**
- 10: $k \leftarrow k \bmod k_{\max} + 1$
- 11: **end if**
- 12: **end while**
- 13: **return** best layout evaluated

14: **function** LOCALSEARCH(y) ▷ compass search, best improvement

- 15: $h \leftarrow 0.05r$
- 16: **while** $h \geq 10^{-3}r$ **do**
- 17: $z^* \leftarrow \arg \min F_p$ over the $4N$ points $y \pm h e_i$ (box-repaired)
- 18: **if** $F_p(z^*) < F_p(y)$ **then**
- 19: $y \leftarrow z^*$
- 20: **else**
- 21: $h \leftarrow h/2$
- 22: **end if**
- 23: **end while**
- 24: **return** y
- 25: **end function**

- rotor radius $R = 38.5$ m and minimum spacing $4D = 8R = 308$ m;
- cut-in speed $s_{\text{cut-in}} = 3.5$ m/s, rated speed $s_{\text{rated}} = 14$ m/s and rated power $P_{\text{rated}} = 1500$ kW;
- linear power-curve parameters $\lambda' = 140.86$ and $\eta = -500$;
- thrust coefficient $C_T = 0.8$ and wake spreading constant $K = 0.075$;
- constraint penalty $\lambda = 10^{10}$ in (7).

B. Methods Compared

We compare the proposed hybrid with six reference methods. All of them are evaluated with the objective (7), the bounds $[-r, r]^{2N}$ and the same budget:

- LX-SSA-VNS (proposed): Algorithm 3 with $\rho = 0.5$, i.e., 50 LX-SSA iterations (3,030 calls) followed by 3,000 calls of basic VNS.
- LX-SSA [30]: Algorithm 2 with $\phi = 0$, $\chi = 1$, population 30 and 100 iterations.
- SSA [26]: the same leader and follower dynamics without the Laplace candidate; population 30 and 200 iterations.

- PSO: inertia weight $w = 0.7$ and acceleration coefficients $c_1 = c_2 = 2$, with unbounded velocities and positions clipped to the bounds; population 30 and 200 iterations.
- DE: DE/rand/1/bin with $F = 0.5$ and $CR = 0.9$; population 30 and 200 iterations.
- VNS: the basic VNS of Mladenović and Hansen [40], [41] in the continuous form of Mladenović et al. [42]. It is exactly Phase 2 of Algorithm 3 (ℓ_∞ shaking of the whole layout with $k_{\max} = 5$ and $\rho_k = 0.1kr$, and a complete best-improvement compass search), started from the best member of the initial population instead of the LX-SSA food source.
- Multistart SLSQP (MS-SLSQP), a gradient-based method: the sequential least-squares quadratic programming solver of SciPy [43] minimizes the wake loss with the spacing and boundary constraints imposed explicitly and forward-difference gradients (1-m step). It starts from the members of the initial population in the order of their F_p values and restarts until the budget is used up.

SSA, LX-SSA, PSO and DE use our original implementation. We implemented VNS, MS-SLSQP and the hybrid for this study; the hybrid calls the original LX-SSA code in its first phase. For the ablation in Section VII-F, we also use SSA-VNS, which replaces LX-SSA by SSA in Phase 1 (100 SSA iterations, 3,030 calls) and is otherwise identical to LX-SSA-VNS.

C. Budget, Seeds and Pairing

Every run has a budget of exactly 6,030 objective calls, counted by a wrapper around the objective function. The count includes the initial population and, for MS-SLSQP, the calls used for the finite-difference gradients. For LX-SSA this corresponds to 100 iterations, because each follower evaluates two candidates before the population is re-evaluated ($30 + 100 \times 60$ calls); SSA, PSO and DE reach the same count in 200 iterations ($30 + 200 \times 30$), and LX-SSA-VNS in $3,030 + 3,000$ calls. Each method is run with the seeds 1–30. Every optimizer seeds the random-number generator and draws its initial population first, so runs with the same seed start from the *same* initial population for all methods, and the runs of a test case are paired by seed.

D. Performance Measures and Statistical Analysis

For each run we record the final objective and the wake loss (as a percentage of the wake-free objective), whether the final layout is feasible (spacing and boundary satisfied within 10^{-6} m), the convergence curve, i.e., the best feasible objective after every 30 calls, and the wall-clock time. Means and standard deviations are computed over the feasible runs.

Within each test case, the 30 seed-paired runs are compared by a run-level Friedman test and by two-sided Wilcoxon signed-rank tests of LX-SSA-VNS against each of the other six methods. The six p values of each case are adjusted with Holm's procedure [48], and the effect sizes are reported as matched-pairs rank-biserial correlations. For the ranking, a feasible run beats an infeasible one, and infeasible runs are compared by their spacing shortfall. Across the 68 cases, the

TABLE II: Case-level Friedman analysis over all 68 benchmark cases (blocks: cases; ranks of the mean feasible objective, 1 = best). Friedman $\chi_F^2 = 168.4$ (6 d.f.), $p = 9.93 \times 10^{-34}$; Iman–Davenport $F_F = 47.1$. Last column: Holm-adjusted p value of the average-rank z test against LX-SSA-VNS (best-ranked method: LX-SSA-VNS); “Best”: cases in which the method alone has the highest mean (remaining cases are ties); “Feas.”: percentage of feasible runs.

Method	Avg. rank	Best	Feas. (%)	p vs. LX-SSA-VNS
LX-SSA-VNS	1.89	31	99.9	–
VNS	2.85	13	100.0	0.010
SSA	3.85	1	97.6	2.33×10^{-7}
PSO	4.14	3	86.0	3.76×10^{-9}
MS-SLSQP	4.44	12	99.9	2.28×10^{-11}
LX-SSA	4.76	0	96.9	4.24×10^{-14}
DE	6.07	0	71.4	1.07×10^{-28}

methods are ranked within each case by their mean feasible objective and compared by a case-level Friedman test with the Iman–Davenport correction and Holm-adjusted post hoc tests on the average ranks, following Demšar [47].

E. Implementation Verification

We vectorized the objective function and checked it against the original implementation; the relative difference is below 10^{-6} on random feasible and infeasible layouts. It also reproduces the objective values of 720 archived final layouts from an earlier run of the original code, with a maximum absolute deviation of 8.7×10^{-11} . Re-running the original SSA, LX-SSA, PSO and DE code with the archived settings reproduces the archived result distributions: 22 of 24 Mann–Whitney comparisons give $p \geq 0.05$, and the means differ by at most 0.21%. This also confirms the Laplace parameters $\phi = 0$ and $\chi = 1$.

F. Computing Environment

All runs were carried out in a Linux container with an Intel Xeon processor (2.80 GHz, 4 virtual cores), four runs in parallel, using Python 3.11.15, NumPy 2.4.6 and SciPy 1.17.1. Each run was timed individually. The scripts, the per-run results (including the final coordinates and convergence curves) and the code that generates the figures are provided with the revision.

VII. NUMERICAL RESULTS

In this section, we report 14,280 runs (68 test cases, seven methods and 30 seed-paired runs, all with 6,030 objective calls). We then present the component ablation of the hybrid, which adds 2,040 runs of SSA-VNS and 720 runs with other budget splits, and the Horns Rev 1 site case. The detailed results of every case are given in Tables IV–IX.

A. Overall Comparison

Table II and Fig. 4 summarize the comparison over all cases. The Friedman test over the 68 cases rejects the hypothesis that the seven methods perform equally ($\chi_F^2 = 168.4$, 6 d.f.,

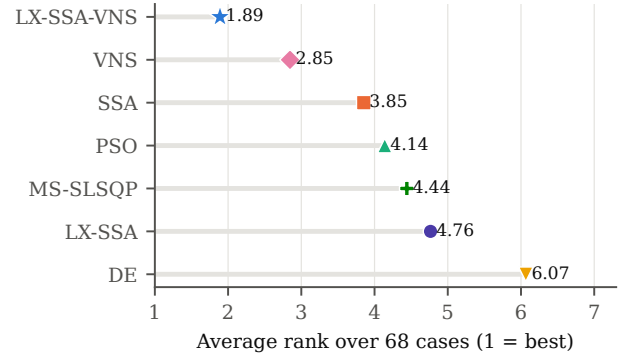


Fig. 4: Average rank of the seven methods over the 68 benchmark cases (ranks of the mean feasible objective within each case; 1 = best).

TABLE III: Pairwise outcome of the hybrid LX-SSA-VNS against each method: number of cases in which LX-SSA-VNS is significantly better / not significantly different / significantly worse (two-sided Wilcoxon signed-rank test on 30 seed-paired runs, Holm-adjusted over the six comparisons of each case, $\alpha = 0.05$).

DS	r (m)	Cases	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
I	500	9	3/6/0	3/6/0	3/6/0	4/5/0	1/7/1	3/3/3
I	750	11	7/4/0	6/5/0	5/6/0	9/2/0	2/9/0	7/4/0
I	1000	14	7/7/0	5/9/0	10/4/0	12/2/0	1/13/0	10/4/0
II	500	9	4/5/0	4/5/0	3/6/0	6/3/0	0/8/1	2/7/0
II	750	11	7/4/0	5/6/0	6/5/0	9/2/0	0/11/0	6/4/1
II	1000	14	11/3/0	9/5/0	9/5/0	12/2/0	2/12/0	8/6/0
All		68	39/29/0	32/36/0	36/32/0	52/16/0	6/60/2	36/28/4

$p = 9.9 \times 10^{-34}$; Iman–Davenport $F_F = 47.1$). LX-SSA-VNS has the best average rank (1.89) and alone attains the highest mean objective in 31 cases. VNS follows with 2.85 (13 cases), and then come SSA (3.85), PSO (4.14), MS-SLSQP (4.44, 12 cases), LX-SSA (4.76) and DE (6.07). In the Holm-adjusted post hoc tests, LX-SSA-VNS ranks significantly better than every other method, including VNS ($p_{\text{Holm}} = 0.010$); for the remaining five methods, $p_{\text{Holm}} \leq 2.3 \times 10^{-7}$.

The run-level tests in Table III show where these differences come from. Against LX-SSA, the hybrid is significantly better in 39 cases and never worse, with rank-biserial correlations between 0.53 and 1.00 (median 0.74). Of the 29 cases without a significant difference, 22 have $N \leq 7$, where LX-SSA already comes close to the best layouts. Against SSA, PSO and DE, the hybrid is significantly better in 32, 36 and 52 cases, respectively, and never worse. MS-SLSQP is beaten in 36 cases but wins four dense cases (Data Set I, 500 m, $N = 8$ –10, and Data Set II, 750 m, $N = 12$), where its explicit handling of the constraints pays off. The comparison with VNS is the closest one: the two methods are statistically indistinguishable in 60 cases, the hybrid is better in six cases with $N = 5$ to 9 (r_{rb} from 0.51 to 0.63), and VNS is better in the two three-turbine cases of the 500-m farm. Hybridization thus moves LX-SSA from the sixth to the first average rank. Its margin over VNS, which forms its own second phase, is small

in individual cases but consistent over the whole benchmark.

B. Solution Quality by Farm Size and Turbine Count

Fig. 5 and Tables IV–IX show how the wake loss depends on the case. It grows with N , falls as the farm becomes larger, and is higher for the broader wind rose of Data Set II. For $N \leq 3$, all methods stay within 0.3% of the wake-free objective. As N grows, the curve of the hybrid separates from that of LX-SSA and joins the curves of VNS and MS-SLSQP. For example, for 15 turbines in the 1000-m farm (Data Set I), the mean wake loss is 1.78% for LX-SSA-VNS, 1.99% for VNS, 2.20% for MS-SLSQP, 2.80% for SSA and 3.32% for LX-SSA; for 12 turbines in the 750-m farm (Data Set II), the corresponding values are 5.95%, 6.17%, 5.60%, 6.79% and 7.34%. In the largest case of every farm, the hybrid reduces the mean wake loss of LX-SSA by 1.4 to 1.9 percentage points. It has the lowest mean wake loss in both 15-turbine cases of the 1000-m farm, while MS-SLSQP has the lowest value in the other four largest cases.

C. Feasibility

Fig. 6 shows the share of runs that end with a feasible layout. LX-SSA-VNS is feasible in 99.9% of all runs; only one run in each of the two 500-m ten-turbine cases fails. The other methods reach 96.9% (LX-SSA), 97.6% (SSA), 86.0% (PSO), 71.4% (DE), 100% (VNS) and 99.9% (MS-SLSQP). In the densest cases (500 m, $N = 10$), LX-SSA is feasible in only 13 of 30 runs, and the complete local search of the VNS phase repairs almost all of the layouts it receives. The feasibility counts of the penalty-based methods are the same for both wind data sets, because as long as every candidate is infeasible, the penalty dominates F_p .

D. Convergence Behavior

Figs. 7 and 8 show the effect of the hybridization directly. Up to 3,030 calls, the hybrid runs a shortened LX-SSA and is therefore behind the full LX-SSA run. After the switch, its curve drops steeply, whereas LX-SSA and SSA stagnate after about 3,000 calls. Over all feasible runs, the VNS phase removes on average 39.0% (median 31.9%) of the wake loss left at the end of the LX-SSA phase, and it turns 118 runs that were still infeasible at the switch into feasible ones. VNS alone reaches low wake losses earlier because it spends its whole budget on refinement, but the hybrid catches up after the switch and ends with a lower mean wake loss in five of the six largest cases. MS-SLSQP is the fastest method to find a feasible layout.

E. Distribution of Results

Fig. 9 shows the spread of the final wake loss over the 30 runs for moderate turbine counts. LX-SSA-VNS has the lowest median in all three cases of Data Set II and the second-lowest in the 500-m case of Data Set I, and its interquartile range is smaller than that of LX-SSA in five of the six cases. DE has the highest median in five cases.

F. Component Ablation

To understand where the improvement of the hybrid comes from, we carry out a component ablation. The hybrid has three ingredients: a swarm phase, the Laplace step inside that swarm phase, and the VNS phase. We therefore compare SSA, LX-SSA, VNS, SSA-VNS and LX-SSA-VNS on all 68 cases with the same seeds and budget (Table X and Fig. 10). The Friedman test over the five variants is highly significant ($\chi_F^2 = 156.5$, 4 d.f., $p = 8.1 \times 10^{-33}$).

Adding the VNS phase is the most important step. LX-SSA-VNS beats LX-SSA in 39 cases and SSA-VNS beats SSA in 40 cases, without a single loss, and the mean wake loss falls by 0.43 and 0.33 percentage points, respectively. Starting VNS from the food source of a swarm, rather than from the best initial point, helps moderately: both two-phase variants have a better average rank than VNS (2.25 and 1.62 against 2.90), LX-SSA-VNS beats VNS in five cases and loses two, and SSA-VNS beats it in ten cases and loses one.

The Laplace step of LX-SSA, on the other hand, does not help. Without VNS, LX-SSA is worse than SSA in seven cases and never better. Inside the hybrid, LX-SSA-VNS and SSA-VNS are statistically indistinguishable in 66 cases, SSA-VNS is better in the remaining two, and SSA-VNS has the better average rank (1.62 against 2.25). The split of the budget between the two phases matters little (Table XI). The 50% split gives the highest mean in eight of the 12 cases, but only one difference is significant (Data Set I, 1000 m, $N = 15$, where the 50% split is better than the 25% split, $p_{\text{Holm}} = 0.024$).

Taken together, the ablation shows that the gain of LX-SSA-VNS comes from combining a swarm exploration phase with a complete VNS refinement phase, and not from the particular Laplace mechanism of LX-SSA, since a standard SSA first phase performs at least as well. The counts against VNS differ slightly from Table III because the Holm correction is applied to a different family of comparisons.

G. Optimized Layouts

Fig. 11 compares, for the largest turbine count of each farm, the best LX-SSA-VNS layout with the best layout found by any method. The best layout overall comes from LX-SSA-VNS in four cases, from MS-SLSQP in one (Data Set I, 500 m) and from VNS in one (Data Set II, 750 m). In all of them, most turbines lie close to the boundary, which maximizes their mutual distances, while the remaining turbines sit in the interior at positions staggered with respect to the dominant wind directions.

H. Computational Cost

At equal numbers of objective calls, all methods need almost the same time: 0.37 ms per call for VNS, 0.39 ms for LX-SSA-VNS, 0.40 ms for LX-SSA and SSA, 0.41 ms for PSO, 0.47 ms for DE and 0.49 ms for MS-SLSQP. The objective evaluation dominates the cost, so neither the Laplace step nor the VNS bookkeeping adds a measurable overhead. One 6,030-call run of LX-SSA-VNS takes 1.7–3.5 s on average, depending on N .

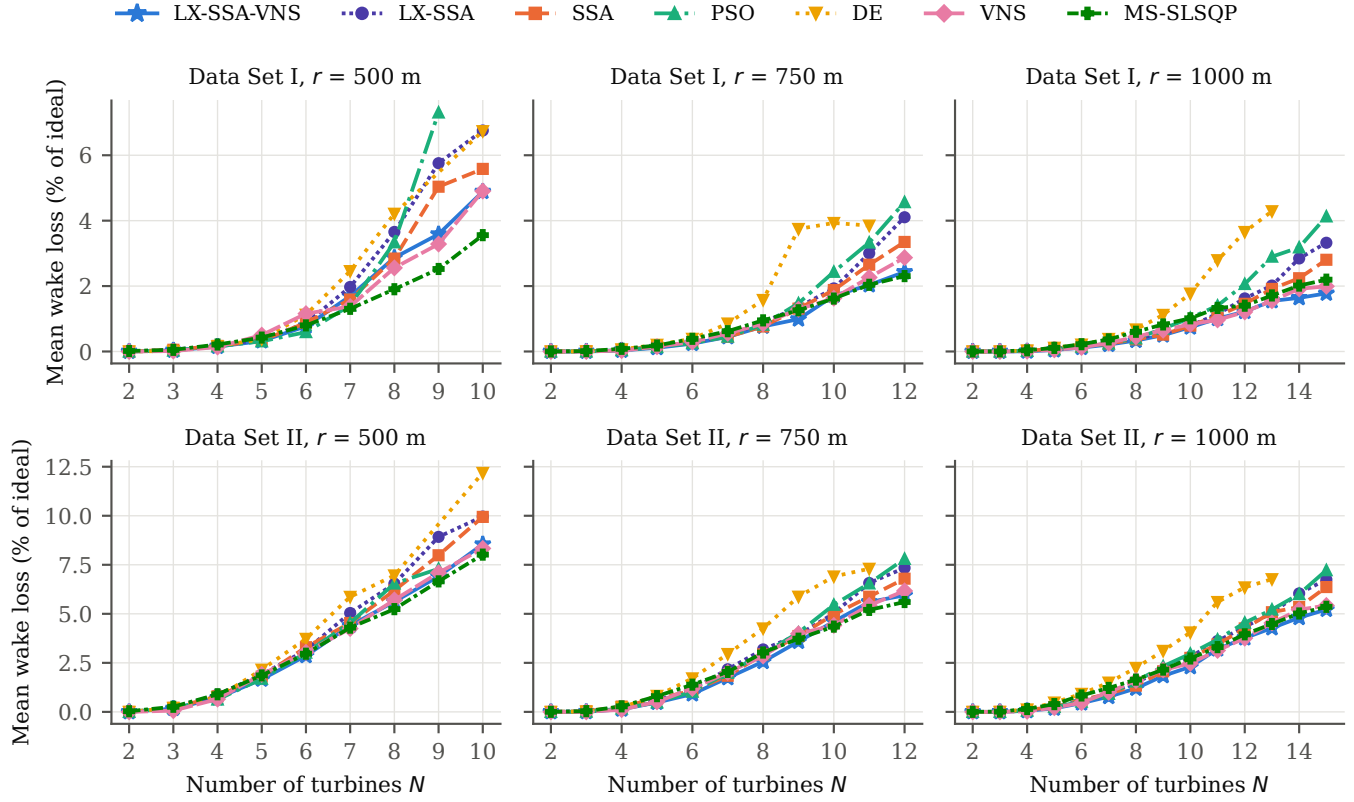


Fig. 5: Mean wake loss (percentage of the wake-free objective) of the feasible runs versus the prescribed number of turbines; a point is omitted when a method has no feasible run.

TABLE IV: Data Set I, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28087.2 (4.6)
3	42137.2	42123.3 (12.8)	42114.8 (14.2)	42114.9 (10.3)	42124.7 (8.5)	42130.7 (8.8)	42131.6 (9.7)	42116.0 (10.1)
4	56182.9	56098.0 (32.4)	56078.8 (22.6)	56086.8 (19.8)	56094.3 (13.3)	56092.8 (29.8)	56106.1 (21.7)	56063.2 (30.9)
5	70228.7	70004.0 (60.7)	69972.6 (72.0)	69994.0 (34.0)	70010.5 (27.2)	69960.5 (78.6)	69876.9 (199.8)	69930.5 (113.0)
6	84274.4	83625.4 (450.2)	83544.9 (465.3)	83535.6 (372.5)	83775.8 (154.1)	83326.7 (761.5)	83299.3 (524.5)	83601.4 (213.1)
7	98320.2	96656.8 (758.8)	96380.6 (1013.8)	96757.0 (692.4)	96935.0 (485.2)	95903.3 (1316.3) ¹⁵	96967.1 (590.6)	97037.0 (484.3)
8	112365.9	109144.5 (1684.4)	108258.3 (1704.3) ²⁸	109172.6 (1362.4)	108616.6 (1480.8) ²⁴	107632.7 (1592.8) ²	109493.5 (2035.2)	110226.6 (585.5)
9	126411.6	121878.0 (1499.7)	119134.2 (2666.4) ²¹	120043.1 (2755.8) ²⁵	117175.2 (7148.7) ²	– ⁰	122259.2 (1232.0)	123222.8 (812.6)²⁸
10	140457.4	133591.9 (1334.5) ²⁹	130967.4 (3191.0) ¹³	132616.3 (2328.0) ¹²	– ⁰	130994.8 (0.0) ¹	133567.5 (1820.9)	135461.0 (1291.9)

TABLE V: Data Set I, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.0 (1.6)
3	42137.2	42137.1 (0.8)	42134.2 (4.8)	42134.4 (4.1)	42137.0 (1.1)	42136.8 (1.5)	42137.2 (0.0)	42132.3 (5.0)
4	56182.9	56165.3 (10.5)	56149.9 (14.0)	56153.8 (17.9)	56160.0 (14.2)	56146.9 (17.5)	56167.0 (13.8)	56137.1 (16.4)
5	70228.7	70148.9 (25.9)	70118.1 (28.6)	70122.2 (26.3)	70135.0 (22.6)	70083.9 (31.7)	70125.8 (47.0)	70100.5 (42.3)
6	84274.4	84071.7 (52.8)	84033.5 (58.3)	84035.2 (58.4)	84059.1 (42.0)	83948.2 (74.1)	84029.2 (141.3)	83947.6 (95.9)
7	98320.2	97881.7 (196.8)	97840.0 (167.7)	97875.2 (169.3)	97872.3 (131.8)	97479.1 (285.1)	97774.4 (232.6)	97710.2 (204.8)
8	112365.9	111504.4 (288.6)	111374.0 (460.8)	111506.5 (334.5)	111448.4 (551.6)	110595.8 (445.6)	111411.4 (493.5)	111296.2 (217.8)
9	126411.6	125161.8 (472.7)	124698.0 (960.4)	124756.2 (610.5)	124550.0 (794.2)	121674.9 (2276.5) ²³	124780.0 (716.8)	124818.4 (466.3)
10	140457.4	138061.5 (775.1)	137748.6 (1116.3)	137817.6 (1160.1)	137036.8 (1051.3) ²⁸	134937.7 (998.5) ⁴	138170.4 (764.7)	138184.7 (595.3)
11	154503.1	151363.4 (1069.4)	149863.8 (1886.6) ²⁹	150404.1 (1042.2) ²⁹	149356.6 (1849.0) ²⁰	148555.7 (1134.8) ³	151004.2 (1339.0)	151352.9 (560.2)
12	168548.8	164450.7 (1126.0)	161630.5 (1988.2) ²⁸	162903.5 (1685.8)	160854.7 (2108.4) ⁷	– ⁰	163712.4 (1575.6)	164663.8 (674.6)

TABLE VI: Data Set I, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.3 (0.7)
3	42137.2	42137.2 (0.0)	42136.7 (1.4)	42136.7 (1.6)	42137.2 (0.0)	42137.2 (0.0)	42137.2 (0.0)	42135.9 (2.6)
4	56182.9	56179.5 (5.3)	56172.9 (7.8)	56169.4 (9.2)	56173.0 (6.8)	56166.9 (8.4)	56177.3 (8.6)	56162.5 (10.5)
5	70228.7	70197.7 (17.0)	70183.3 (21.0)	70174.0 (19.7)	70182.6 (13.0)	70148.7 (22.4)	70193.3 (24.5)	70151.4 (32.7)
6	84274.4	84175.0 (31.8)	84142.7 (23.5)	84149.9 (29.7)	84152.3 (22.2)	84096.1 (24.0)	84170.0 (49.5)	84087.4 (92.6)
7	98320.2	98111.9 (67.7)	98078.4 (82.1)	98084.7 (47.0)	98083.0 (35.8)	97942.3 (67.9)	98064.6 (118.7)	97955.8 (136.6)
8	112365.9	111990.4 (150.7)	111924.6 (164.7)	111947.1 (125.7)	111915.5 (125.3)	111608.3 (168.9)	111862.5 (246.1)	111688.0 (181.9)
9	126411.6	125781.8 (211.3)	125701.5 (265.4)	125761.1 (168.4)	125531.1 (324.5)	124996.3 (282.9)	125589.7 (312.6)	125366.6 (282.7)
10	140457.4	139414.7 (298.8)	139354.2 (388.5)	139348.0 (433.0)	139018.6 (478.6)	137969.0 (623.0)	139290.3 (447.7)	139031.6 (341.6)
11	154503.1	152951.9 (538.1)	152736.8 (558.8)	152825.1 (385.8)	152337.8 (768.8) ²⁹	150189.7 (1068.9) ²⁹	153010.5 (393.8)	152445.9 (402.0)
12	168548.8	166492.7 (656.7)	165814.4 (827.9)	166091.9 (816.6)	165072.5 (1367.5) ²⁸	162390.4 (1765.9) ¹⁷	166510.6 (669.8)	166188.4 (681.0)
13	182594.6	179771.9 (786.7)	178913.6 (967.4)	179101.5 (1028.9)	177302.0 (2352.6) ²²	174763.4 (1389.6) ⁴	179762.6 (1091.2)	179465.2 (506.2)
14	196640.3	193414.7 (790.2)	191046.6 (1754.6)	192235.7 (1504.8)	190385.1 (1544.4) ¹⁸	– ⁰	192874.6 (1745.6)	192677.8 (595.4)
15	210686.1	206943.1 (740.2)	203681.8 (1899.5) ²⁹	204777.6 (1420.5)	201985.5 (1678.3) ⁹	– ⁰	206486.7 (1093.6)	206059.6 (761.8)

TABLE VII: Data Set II, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14626.0 (5.6)
3	21946.1	21909.0 (14.7)	21899.7 (9.5)	21900.9 (11.1)	21901.5 (6.3)	21901.2 (5.5)	21932.3 (17.2)	21888.8 (45.5)
4	29261.5	29055.1 (84.8)	29004.2 (87.4)	29028.7 (93.8)	29077.9 (64.5)	29044.2 (61.5)	29070.8 (94.1)	28998.9 (104.2)
5	36576.9	35969.8 (125.2)	35880.1 (168.6)	35918.9 (74.0)	35959.4 (119.0)	35789.6 (193.8)	35878.2 (160.9)	35896.9 (110.0)
6	43892.3	42636.3 (185.8)	42512.1 (227.9)	42444.3 (224.5)	42568.5 (159.7)	42257.5 (374.9)	42560.3 (202.5)	42590.7 (236.0)
7	51207.6	48955.1 (239.9)	48626.5 (340.5)	48887.9 (304.1)	48885.9 (278.3)	48203.9 (767.8) ¹⁵	49032.1 (314.4)	49007.9 (208.2)
8	58523.0	55234.6 (495.0)	54699.5 (620.7) ²⁸	54900.5 (471.3)	54693.1 (404.3) ²⁴	54451.3 (618.7) ²	55182.8 (738.6)	55457.2 (287.3)
9	65838.4	61289.5 (782.7)	59963.3 (652.2) ²¹	60581.0 (844.8) ²⁵	61035.6 (137.9) ²	– ⁰	61166.8 (574.8)	61457.0 (370.7)
10	73153.8	66900.5 (785.8) ²⁹	65868.7 (586.0) ¹³	65880.7 (761.6) ¹²	– ⁰	64240.2 (0.0) ¹	67056.0 (749.1)	67288.3 (300.1)²⁹

TABLE VIII: Data Set II, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.4 (1.5)
3	21946.1	21945.0 (3.7)	21937.3 (8.1)	21940.0 (8.2)	21943.2 (5.4)	21942.5 (5.8)	21946.1 (0.0)	21938.2 (12.2)
4	29261.5	29222.7 (25.5)	29201.4 (25.7)	29182.5 (36.2)	29213.0 (20.4)	29195.3 (26.2)	29227.3 (29.9)	29179.0 (59.2)
5	36576.9	36401.8 (88.0)	36366.8 (74.3)	36332.0 (91.4)	36400.0 (62.2)	36277.7 (105.9)	36389.7 (97.1)	36283.6 (98.0)
6	43892.3	43493.2 (148.6)	43377.4 (175.7)	43419.5 (153.1)	43441.8 (107.5)	43142.2 (144.8)	43367.4 (201.2)	43283.1 (153.3)
7	51207.6	50319.4 (178.7)	50095.4 (193.6)	50264.8 (202.5)	50188.2 (199.5)	49696.6 (156.3)	50180.2 (303.8)	50164.7 (205.9)
8	58523.0	57024.1 (299.7)	56659.2 (332.0)	56806.2 (303.6)	56786.4 (311.8)	56033.1 (331.4)	56855.9 (391.5)	56752.4 (256.7)
9	65838.4	63476.9 (358.5)	63252.7 (342.8)	63291.6 (402.5)	63180.2 (411.7)	61970.8 (763.0) ²³	63217.0 (515.4)	63381.6 (322.9)
10	73153.8	69786.4 (553.5)	69480.7 (385.7)	69520.3 (512.6)	69159.6 (436.1) ²⁸	68103.2 (423.7) ⁴	69930.2 (355.0)	69972.9 (458.1)
11	80469.2	75963.7 (535.8)	75181.6 (681.1) ²⁹	75742.5 (677.1) ²⁹	75199.6 (585.7) ²⁰	74596.6 (476.3) ³	76099.7 (566.3)	76284.4 (433.9)
12	87784.5	82564.5 (503.5)	81341.8 (792.1) ²⁸	81822.1 (625.7)	80936.3 (798.4) ⁷	– ⁰	82368.6 (603.1)	82867.2 (399.2)

I. Horns Rev 1 Site Case

To test the methods closer to a real project, we modeled the Horns Rev 1 offshore farm with its real turbine, wind climate and site outline, all taken from the PyWake distribution [44]. The model contains the installed layout of 80 Vestas V80 turbines (rotor diameter 80 m, 2 MW) with the V80 power and thrust curves, the measured 12-sector wind climate, and the parallelogram spanned by the installed turbines as the site boundary. We keep the Jensen top-hat wake, with $k = 0.04$ as commonly used offshore, and root-sum-square superposition,

and we use the tabulated power curve with a 25 m/s cut-out, a speed-dependent thrust coefficient at the free-stream speed, 5° direction bins and 1 m/s speed bins. For the installed layout, our model gives 664.6 GWh/yr and a wake loss of 10.95%, against 662.5 GWh/yr and 10.96% from PyWake’s Jensen model with the same settings, a difference below 0.3%. We optimized the 16-turbine block that PyWake uses as an example (30 seeds) and the complete 80-turbine farm (10 seeds) with all seven methods, 6,030 calls and the 4D (320 m) minimum spacing.

TABLE IX: Data Set II, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “—” means no feasible run.

N	Ideal	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.5 (1.0)
3	21946.1	21946.1 (0.0)	21945.3 (2.8)	21945.6 (1.8)	21945.8 (2.0)	21946.1 (0.0)	21946.1 (0.0)	21943.5 (4.8)
4	29261.5	29245.9 (15.1)	29228.0 (14.9)	29232.5 (18.5)	29243.3 (13.6)	29225.9 (12.1)	29253.1 (11.7)	29218.7 (24.2)
5	36576.9	36510.9 (37.7)	36468.2 (46.1)	36485.9 (37.5)	36486.0 (36.0)	36402.8 (39.4)	36498.4 (49.1)	36432.0 (50.4)
6	43892.3	43696.6 (69.9)	43656.6 (73.3)	43626.4 (74.5)	43665.8 (65.6)	43488.5 (69.7)	43686.1 (101.2)	43524.8 (86.8)
7	51207.6	50810.8 (104.4)	50727.3 (119.3)	50731.8 (118.8)	50739.2 (134.6)	50442.7 (136.0)	50696.0 (203.8)	50582.5 (165.5)
8	58523.0	57822.5 (199.4)	57645.1 (186.6)	57739.3 (142.0)	57616.2 (157.7)	57210.0 (191.8)	57588.6 (311.3)	57560.8 (206.6)
9	65838.4	64629.4 (340.1)	64428.7 (273.1)	64476.4 (203.0)	64308.4 (234.8)	63790.1 (227.2)	64430.8 (311.5)	64419.1 (268.8)
10	73153.8	71474.6 (373.6)	71099.4 (266.9)	71139.4 (336.8)	70985.2 (314.4)	70190.1 (286.9)	71342.3 (339.4)	71161.4 (275.8)
11	80469.2	77892.9 (457.0)	77561.6 (498.0)	77697.5 (425.8)	77501.2 (352.1) ²⁹	75960.9 (577.6) ²⁹	77948.1 (483.3)	77812.4 (323.7)
12	87784.5	84480.4 (528.2)	83981.2 (515.0)	84049.6 (660.3)	83795.7 (565.1) ²⁸	82211.6 (905.2) ¹⁷	84485.7 (606.8)	84312.7 (466.3)
13	95099.9	91033.2 (568.8)	90303.5 (674.2)	90267.0 (482.6)	90139.1 (533.9) ²²	88665.1 (496.0) ⁴	90787.8 (806.9)	90837.9 (572.0)
14	102415.3	97496.3 (644.9)	96229.6 (1019.3)	96936.5 (669.3)	96266.2 (730.7) ¹⁸	— ⁰	97079.1 (980.3)	97276.4 (405.3)
15	109730.7	104002.4 (617.0)	102327.4 (838.7) ²⁹	102746.3 (827.8)	101804.8 (571.6) ⁹	— ⁰	103777.6 (794.4)	103853.3 (475.2)

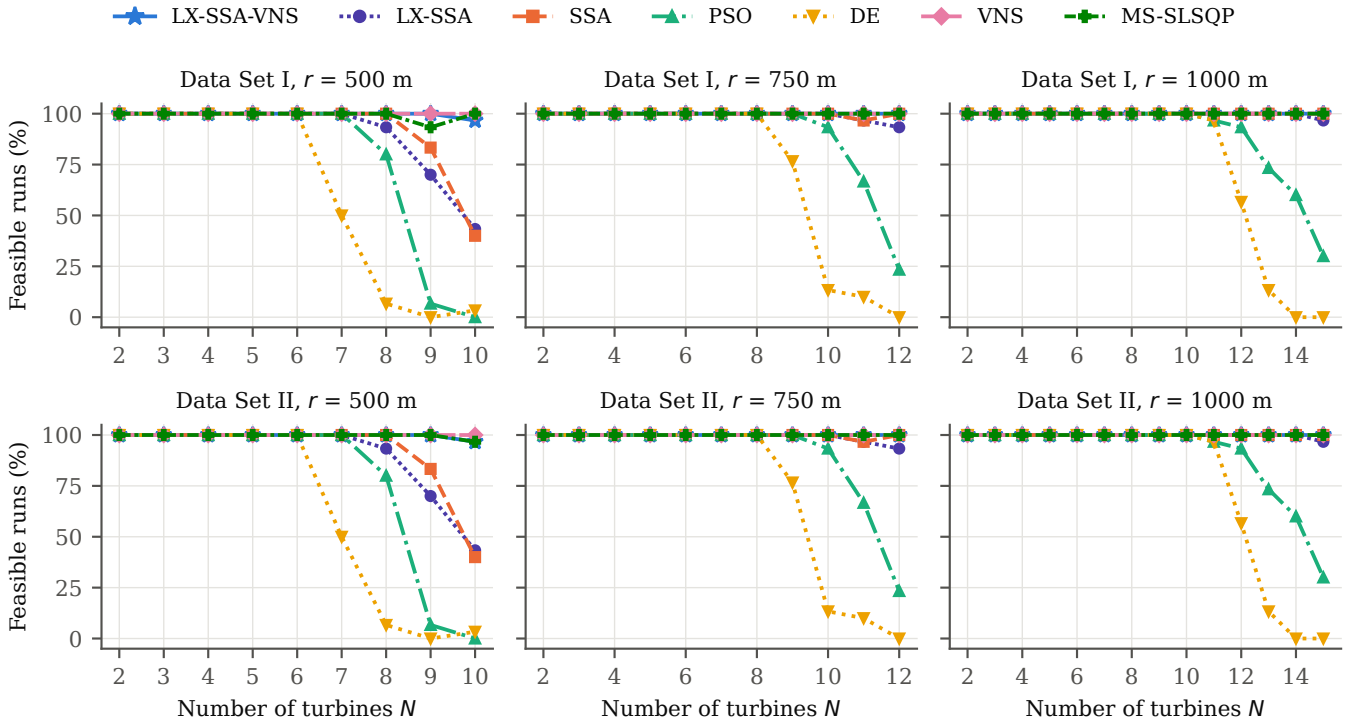


Fig. 6: Percentage of the 30 runs that end with a feasible layout (all turbines inside the farm and at least 308 m apart).

The results are given in Table XII and Figs. 12 and 13. In the 16-turbine block, LX-SSA-VNS attains the highest mean AEP (137.77 GWh/yr) and is feasible in all 30 runs. It is significantly better than LX-SSA ($p_{\text{Holm}} = 5.7 \times 10^{-6}$), SSA ($p_{\text{Holm}} = 3.1 \times 10^{-4}$) and MS-SLSQP ($p_{\text{Holm}} = 2.4 \times 10^{-4}$), while its difference from VNS (137.67 GWh/yr) is not significant ($p_{\text{Holm}} = 0.685$). PSO and DE never find a feasible layout inside the parallelogram, and none of the methods improves on the installed 7D grid, which reaches 139.51 GWh/yr. The full farm is a much harder problem. With 160 variables and 3,160 spacing constraints, no run of LX-SSA-VNS, LX-SSA, SSA, PSO, DE or VNS ends with a feasible layout. MS-SLSQP,

which handles the constraints explicitly, is feasible in all ten runs, but its mean AEP (653.9 GWh/yr) remains 1.6% below that of the installed layout. Realistic farm sizes therefore call for larger budgets, feasible initialization or explicit constraint handling, and a regular installed layout is a strong reference.

VIII. SENSITIVITY TO THE MODELING ASSUMPTIONS

A. Power Curve and Wake Model

To see how far our conclusions depend on the benchmark power curve and wake model, we re-evaluated every feasible final layout, without re-optimization, with three alternative models:

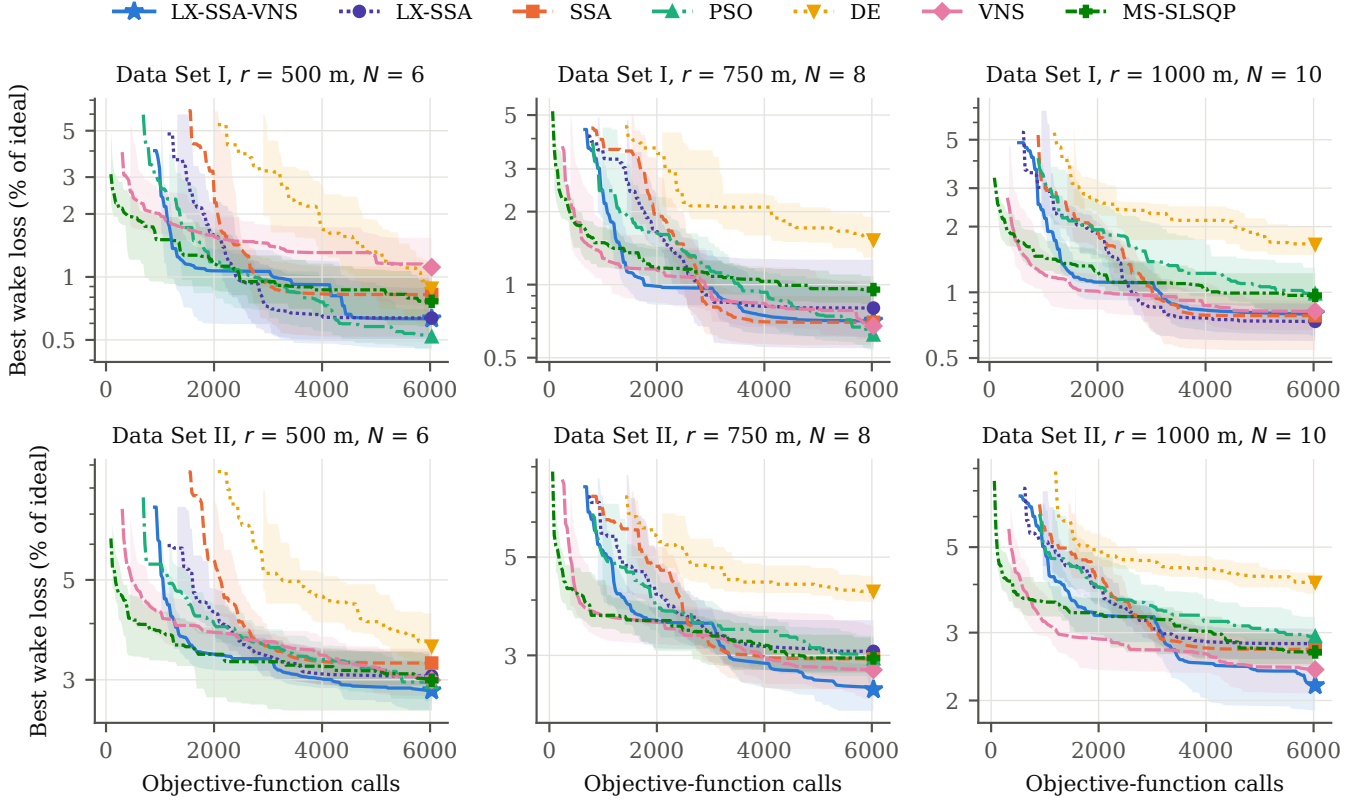


Fig. 7: Convergence for moderate turbine counts: median (line) and interquartile range (band) of the best feasible wake loss found so far over the 30 runs, versus the number of objective calls (logarithmic scale). A curve starts once at least half of the runs have found a feasible layout; the LX-SSA-VNS curve switches from the LX-SSA to the VNS phase at 3,030 calls.

TABLE X: Component ablation over the 68 benchmark cases (30 seed-paired runs, 6,030 calls). Top: phase-1 and phase-2 components, average rank of the mean feasible objective among the five variants (Friedman $\chi^2_F = 156.5$, 4 d.f., $p = 8.10 \times 10^{-33}$) and percentage of feasible runs. Bottom: planned contrasts, number of cases in which the first variant is significantly better / not different / worse (Wilcoxon signed-rank, Holm-adjusted over the six contrasts of each case, $\alpha = 0.05$), and mean difference of the wake loss $\Delta \bar{L}$ (percentage points; negative = first variant better).

Variant	Phase 1	Phase 2	Avg. rank	Feas. (%)
SSA-VNS	SSA	VNS	1.62	99.6
LX-SSA-VNS	LX-SSA	VNS	2.25	99.9
VNS	–	VNS	2.90	100.0
SSA	SSA	–	3.74	97.6
LX-SSA	LX-SSA	–	4.49	96.9

Contrast	Isolates	W/T/L	$\Delta \bar{L}$
LX-SSA-VNS vs. LX-SSA	VNS phase (after LX-SSA)	39/29/0	−0.43
LX-SSA-VNS vs. VNS	LX-SSA start vs. best initial point	5/61/2	−0.07
LX-SSA-VNS vs. SSA-VNS	Laplace step inside the hybrid	0/66/2	+0.06
SSA-VNS vs. SSA	VNS phase (after SSA)	40/28/0	−0.33
SSA-VNS vs. VNS	SSA start vs. best initial point	10/57/1	−0.13
LX-SSA vs. SSA	Laplace step alone	0/61/7	+0.17

TABLE XI: Sensitivity of LX-SSA-VNS to the budget split between the LX-SSA and VNS phases (25%, 50% and 75% of the 6,030 calls for LX-SSA): mean benchmark objective of the feasible runs (superscript: feasible runs when fewer than 30) and Holm-adjusted Wilcoxon signed-rank p of the 50% split against the 25% and 75% splits.

DS	r	N	25%	50%	75%	p (25)	p (75)
1	500	6	83452.8	83625.4	83669.0	0.176	0.919
1	500	10	133540.4 ²⁷	133591.9 ²⁹	133116.1 ²⁴	0.670	0.090
1	750	8	111470.6	111504.4	111630.8	0.777	0.248
1	750	12	163886.1	164450.7	163757.8	0.304	0.304
1	1000	10	139503.7	139414.7	139492.8	0.339	0.328
1	1000	15	206280.9	206943.1	206371.9	0.024	0.067
2	500	6	42609.4	42636.3	42494.4	0.612	0.095
2	500	10	66948.7 ²⁷	66900.5 ²⁹	66693.4 ²⁴	0.855	0.095
2	750	8	56897.0	57024.1	56996.8	0.457	0.777
2	750	12	82420.0	82564.5	82520.3 ²⁹	0.856	0.856
2	1000	10	71406.8	71474.6	71323.6	0.339	0.238
2	1000	15	103752.9	104002.4	103737.0	0.316	0.316

- 1) a cubic power curve [45], $f_c(s) = P_{\text{rated}}(s^3 - s_{\text{cut-in}}^3)/(s_{\text{rated}}^3 - s_{\text{cut-in}}^3)$ between the cut-in and rated speeds;
- 2) the same curve with a 25 m/s cut-out;
- 3) the Gaussian wake model of Bastankhah and Porté-

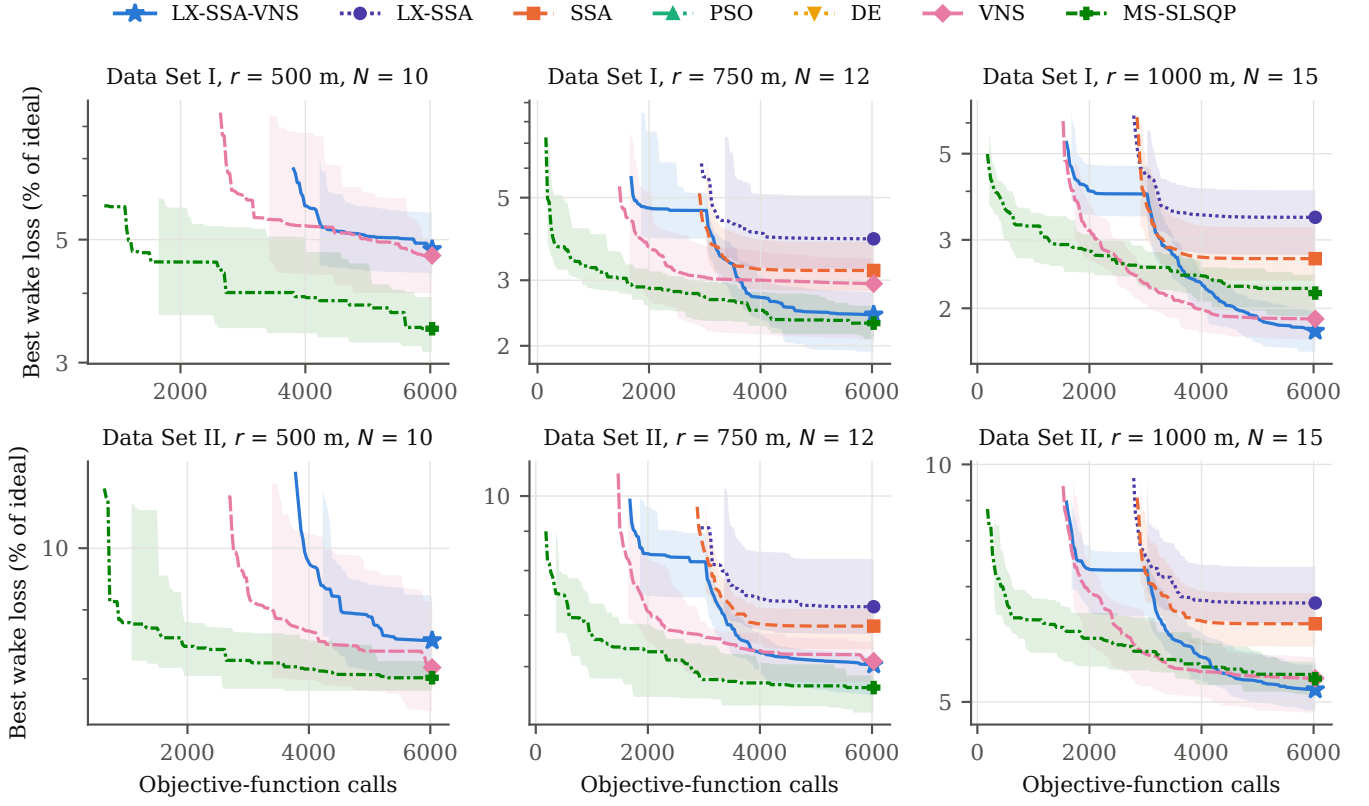


Fig. 8: Convergence for the largest turbine count of each farm (as Fig. 7). Methods for which fewer than half of the runs find a feasible layout have no curve.

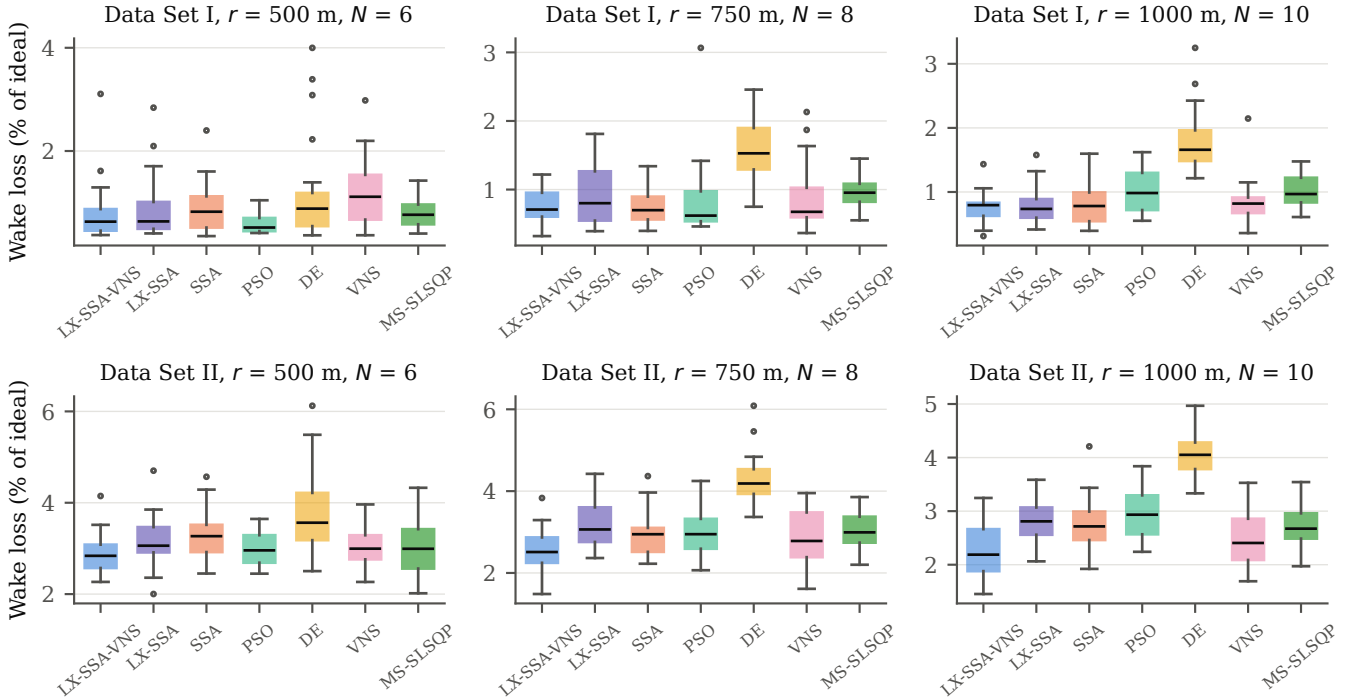


Fig. 9: Distribution of the final wake loss of the feasible runs for moderate turbine counts (30 runs per method).

Agel [46],

$$\delta_{ij} = \left(1 - \sqrt{1 - C_T/(8\sigma^2/D^2)}\right) \exp(-r_{\perp}^2/(2\sigma^2)),$$

$$\sigma/D = k^*x/D + 0.2\sqrt{\beta},$$

with $\beta = \frac{1}{2}(1 + \sqrt{1 - C_T})/\sqrt{1 - C_T}$ and $k^* = 0.04$,

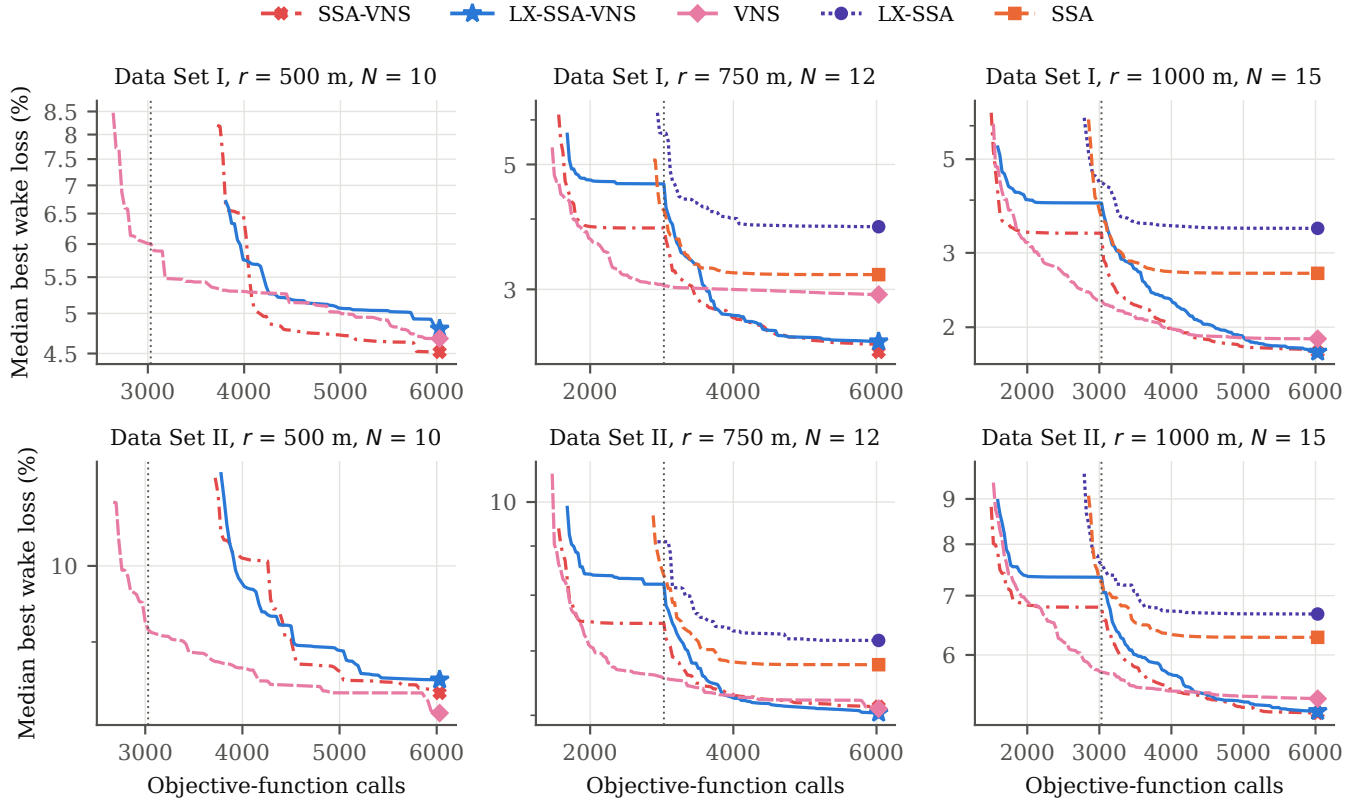


Fig. 10: Component ablation, largest turbine count of each farm: median best feasible wake loss over 30 runs versus objective calls for SSA, LX-SSA, VNS and the two-phase variants SSA-VNS and LX-SSA-VNS (dotted vertical line: switch from the swarm phase to VNS at 3,030 calls).

TABLE XII: Horns Rev 1 site case (real Vestas V80 power and thrust curves, measured 12-sector wind climate, installed farm outline; Jensen wake with $k = 0.04$; minimum spacing $4D = 320$ m): AEP in GWh/yr, mean (SD) over the feasible runs at 6,030 objective calls (30 seeds for the 16-turbine block, 10 for the 80-turbine farm), mean wake loss relative to the wake-free AEP (“Loss”, %), feasible runs, and Holm-adjusted Wilcoxon signed-rank p of LX-SSA-VNS vs. each method (16-turbine block; run-level Friedman $p = 5.34 \times 10^{-27}$).

Layout / method	16-turbine block (wake-free 149.26 GWh/yr)				80-turbine farm (wake-free 746.3 GWh/yr)		
	AEP	Loss	Feas.	p_{Holm}	AEP	Loss	Feas.
Installed layout	139.51	6.53	—	—	664.6	10.95	—
LX-SSA-VNS	137.77 (0.91)	7.70	30/30	—	—	—	0/10
LX-SSA	136.46 (0.90)	8.58	28/30	5.68×10^{-6}	—	—	0/10
SSA	136.66 (1.00)	8.44	30/30	3.06×10^{-4}	—	—	0/10
PSO	—	—	0/30	1.12×10^{-8}	—	—	0/10
DE	—	—	0/30	1.12×10^{-8}	—	—	0/10
VNS	137.67 (1.02)	7.77	30/30	0.685	—	—	0/10
MS-SLSQP	136.80 (0.77)	8.35	30/30	2.37×10^{-4}	653.9 (2.6)	12.39	10/10

evaluated at the rotor center with the same superposition and Weibull scaling as the benchmark.

Table XIII shows two kinds of sensitivity. First, the absolute energy depends strongly on the model: the linear benchmark curve overstates the expected power relative to the cubic curve by 17.7% for Data Set I and 36.9% for Data Set II (21.5% and 37.1% with the cut-out). The energy values in this paper should therefore be read as benchmark values and not as AEP predictions for a specific turbine. Second, the ranking of the algorithms is stable under the cubic curve (mean Kendall $\bar{\tau} =$

0.97), with LX-SSA-VNS (1.95) and VNS (2.85) still the two best methods, but it changes more under the Gaussian model ($\bar{\tau} = 0.54$; the best method is the same in 43% of the cases), because layouts optimized for the sharp Jensen wake cone are judged differently by a smooth Gaussian deficit. LX-SSA-VNS nevertheless keeps the best average rank under the Gaussian model (2.82), followed by PSO (3.16), MS-SLSQP (3.43) and VNS (3.76).

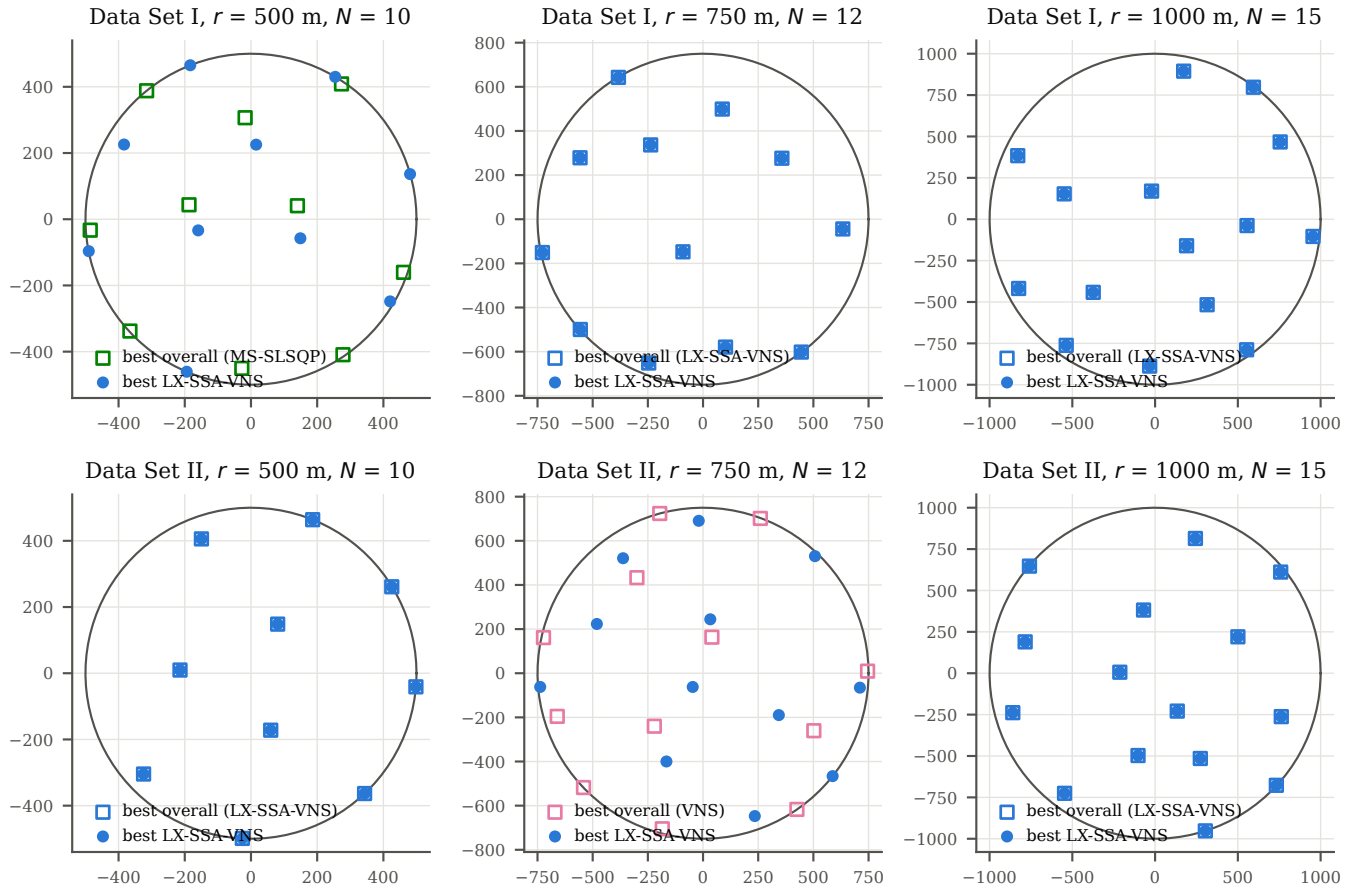


Fig. 11: Best layouts for the largest turbine count of each farm: best LX-SSA-VNS run (filled markers) and best run of any method (open squares, method in the legend). Circle: farm boundary. Coordinates are listed in Appendix A.

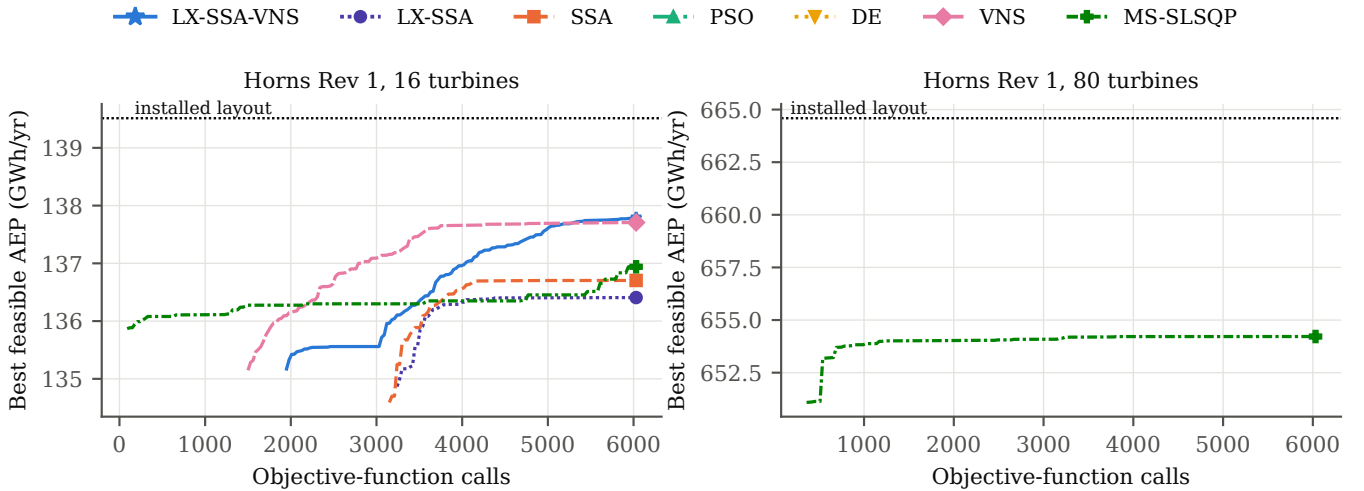


Fig. 12: Horns Rev 1 site case: median best feasible AEP versus objective calls (16-turbine block: 30 runs; 80-turbine farm: 10 runs); dotted line: installed layout. Only MS-SLSQP finds feasible layouts for the 80-turbine farm.

B. Minimum Spacing

The 4D spacing is inherited from the benchmark literature, and we examined its influence in three ways. First, Table XIV gives the largest turbine count for which a multi-start packing

search constructed a feasible layout. These are constructive lower bounds, and the 500-m values agree with the known optimal circle-packing results. With 5D spacing, the 500-m farm holds at most 8 turbines and with 6D only 7, so the

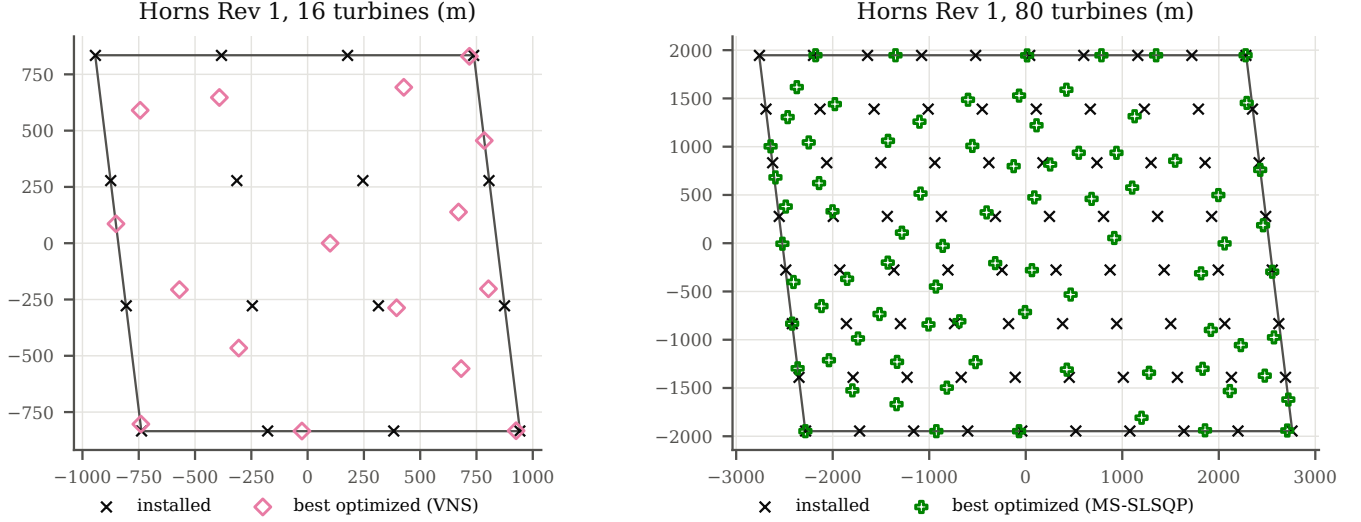


Fig. 13: Horns Rev 1: installed layout (crosses) and best optimized layout (open markers) inside the installed farm outline; local coordinates in metres.

TABLE XIII: Robustness of the results to the benchmark model. All feasible final layouts are re-evaluated (not re-optimized) with alternative power-curve and wake models. Δ : mean relative change of the objective (expected power) with respect to the benchmark model; average rank of each method over the 68 cases (methods with at least five feasible runs); $\bar{\tau}$: mean Kendall rank correlation between the benchmark and alternative orderings of the methods within a case; “Same best”: percentage of cases in which the best method is unchanged.

Model	Δ (%)		Average rank							$\bar{\tau}$	Same best (%)
	DS I	DS II	LX-SSA-VNS	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP		
Benchmark (linear curve, Jensen)	+0.0	+0.0	1.88	4.71	4.22	4.10	5.98	2.79	4.34	—	—
Cubic power curve	-17.7	-36.9	1.95	4.66	4.19	3.99	5.95	2.85	4.41	0.97	97
Cubic curve with 25 m/s cut-out	-21.5	-37.1	1.96	4.66	4.21	3.99	5.96	2.85	4.37	0.97	96
Gaussian wake ($k^* = 0.04$)	-0.2	-0.4	2.82	4.88	4.06	3.16	5.88	3.76	3.43	0.54	43

TABLE XIV: Largest turbine count for which a layout satisfying the circular boundary and the minimum spacing was constructed (multi-start packing search; a constructive lower bound on geometric capacity), compared with the largest tested N .

r (m)	4D (308 m)	5D (385 m)	6D (462 m)	Largest tested N
500	13	8	7	10
750	26	19	13	12
1000	42	29	21	15

500-m cases with $N \geq 9$ (at 5D) or $N \geq 8$ (at 6D) would be infeasible.

Second, the optimized layouts often lie on the 4D constraint. Of the 13,293 feasible final layouts of the seven methods, 48.6% also satisfy 5D and 35.8% satisfy 6D; for $N = 6$ –10 these shares fall to 30.1% and 11.6%, and for $N = 11$ –15 to 3.0% and 0.1%. Layouts optimized for 4D therefore cannot be transferred to a site with larger spacing requirements without re-optimization.

Third, we re-optimized six representative cases with LX-SSA and SSA at 5D and 6D (Table XV). In the 750-m eight-turbine cases, the mean LX-SSA objective falls from 4D to 6D

by 0.95% (Data Set I) and 0.22% (Data Set II), and the wake loss rises from 0.88% to 1.85% and from 3.18% to 3.47%. The spacing rule does not reverse the comparison between LX-SSA and SSA.

C. Boundary Handling

All methods clip the coordinates to the bounding square and penalize turbines outside the circle (Section II-C). An otherwise identical SSA that instead projects every turbine radially onto the circle attains higher mean objectives than the original SSA in six representative cases at 3,030 calls, by 26 to 649 objective units (Mann–Whitney $p < 10^{-4}$ in every case). The boundary rule therefore has a noticeable effect, and we keep it fixed for all methods.

IX. COMPUTATIONAL COMPLEXITY

Let N_t be the number of turbines, N_θ the number of wind-direction bins and N_s the number of wind-speed bins. A direct wake evaluation covers the pairwise turbine interactions for every direction and speed bin, so one objective evaluation costs

$$C_{\text{obj}} = \mathcal{O}(N_t^2 N_\theta N_s),$$

and a run with a budget of B objective calls costs $\mathcal{O}(B N_t^2 N_\theta N_s)$ for every method considered here, including

TABLE XV: Minimum-spacing sensitivity with the authors' LX-SSA and SSA code (6,030 calls, 30 seed-paired runs per cell): mean benchmark objective, Wilcoxon signed-rank p for LX-SSA vs. SSA, and the mean LX-SSA wake loss as a percentage of the ideal objective at 4D and 6D. A superscript gives the number of feasible runs when fewer than 30.

DS	Radius	N	4D (308 m)			5D (385 m)			6D (462 m)			LX-SSA loss (%)
			LX-SSA	SSA	p	LX-SSA	SSA	p	LX-SSA	SSA	p	
1	500	4	56078.8	56086.8	0.237	56082.4	56085.2	0.700	56075.5	56083.9	0.839	0.19 → 0.19
1	750	8	111374.0	111506.5	0.245	111154.1	111163.4	0.871	110314.4 ²⁸	110847.5	0.021	0.88 → 1.85
1	1000	8	111924.6	111947.1	0.887	111953.5	111946.4	0.777	111781.5	111964.7	0.005	0.39 → 0.52
2	500	4	29004.2	29028.7	0.146	28999.5	29019.6	0.229	28981.8	28971.1	0.612	0.88 → 0.96
2	750	8	56659.2	56806.2	0.061	56621.0	56673.3	0.584	56533.9 ²⁸	56690.6	0.038	3.18 → 3.47
2	1000	8	57645.1	57739.3	0.084	57580.7	57655.7	0.140	57475.5	57587.2	0.114	1.50 → 1.79

both phases of LX-SSA-VNS. Box repair is $\mathcal{O}(N_t)$ per candidate and the spacing check $\mathcal{O}(N_t^2)$, and the Laplace draw of LX-SSA, the shaking step of VNS and each compass move are $\mathcal{O}(N_t)$. All of these are dominated by the wake evaluation, which agrees with the almost identical measured cost per call (Section VII-H).

X. LIMITATIONS

Our results should be read within the following limits. The ablation shows that the gain of LX-SSA-VNS comes from the combination of a swarm phase with a complete VNS phase, and not from the Laplace step of LX-SSA: SSA-VNS is statistically equivalent in 66 of the 68 cases, better in two, and has a slightly better average rank. The hybrid is significantly better than VNS over the whole benchmark, but in single cases the two differ significantly in only eight of the 68 cases.

All methods were compared at 6,030 objective calls, and larger budgets could change the ranking; for an 80-turbine farm, none of the penalty-based methods, including the hybrid, found a feasible layout within this budget. The LX-SSA parameters ($\phi = 0$, $\chi = 1$), the VNS settings and the split $\rho = 0.5$ are fixed defaults and were not tuned, and pseudo-gradient and surrogate-based methods are discussed but not benchmarked.

The Jensen model and the linear power curve are benchmark choices. The absolute energy depends strongly on them, the ranking of the methods changes under a Gaussian wake model, and we did not re-optimize the layouts with the alternative models. The 4D spacing and the boundary rule also affect the absolute results. Finally, N is prescribed in every run, and terrain, mixed turbine types, collector cables, grid connection, noise and environmental constraints, and economic objectives are outside the scope of this study.

XI. CONCLUSION

This work presents LX-SSA-VNS, a two-phase hybrid for the continuous wind farm layout problem in which the Laplacian Salp Swarm Algorithm [30] explores the layout space and the basic variable neighborhood search then refines the best layout found. We evaluated the hybrid on the Jensen-Weibull benchmark with explicit constraint handling against LX-SSA, SSA, PSO, DE, VNS and a multistart SLSQP solver on 68 cases (14,280 seed-paired runs at equal budgets), in a component ablation, and on the Horns Rev 1 offshore farm.

LX-SSA-VNS obtained the best average rank (1.89), was significantly better than every other method over the whole benchmark, including VNS ($p_{\text{Holm}} = 0.010$), and returned a feasible layout in 99.9% of the runs. Compared with LX-SSA alone, it was significantly better in 39 of the 68 cases and never worse. The ablation showed that this gain comes from the two-phase design: the VNS phase removes on average 39% of the wake loss left by the swarm, and starting VNS from a swarm solution is better than starting it from the best initial point. The Laplace step, however, does not contribute, since an SSA first phase performs at least as well. At the Horns Rev 1 site, the hybrid gave the highest mean energy yield for a 16-turbine block, although no method improved on the installed layout within the budget and only the constraint-aware SLSQP solver found feasible layouts for the full 80-turbine farm. The ranking of the methods was stable under a cubic power curve but changed considerably under a Gaussian wake model, where the hybrid nevertheless kept the best average rank.

Despite these results, the hybrid shares the limits of penalty-based search on large farms, and its evaluation is tied to a benchmark wake model. In future work, we will study a switch between the two phases that is triggered by the stagnation of the swarm, investigate which properties of the swarm phase make it a good starting point for VNS, and extend the approach with feasibility-preserving initialization, larger budgets for realistic farm sizes, and higher-fidelity wake and power models.

APPENDIX A

COORDINATES OF THE BEST LAYOUTS

Table XVI lists the turbine coordinates (in m, with the farm center at the origin) of the best layout found by any method for the largest turbine count of each farm (Fig. 11). The final coordinates and convergence curves of all runs and the Horns Rev 1 layouts are provided as machine-readable files.

Compliance with Ethical Standards

Conflict of Interest: The authors declare that they have no conflict of interest.

Ethical Approval: This article does not contain any studies with human participants or animals performed by any of the authors.

Informed Consent: Not applicable.

TABLE XVI: Coordinates of the best layout for the largest turbine count of each farm (method and benchmark objective given).

DS	r (m)	N	Method	Objective	Coordinates (ρ_i, σ_i) (m)
1	500	10	MS-SLSQP	138171.2	(-315.2, 388.1); (277.9, -409.2); (-187.2, 43.6); (-27.6, -450.5) (-17.7, 306.6); (460.7, -160.4); (-365.7, -337.8); (-485.5, -33.0) (273.5, 408.8); (140.3, 40.7)
1	750	12	LX-SSA-VNS	166277.4	(-556.0, -499.5); (445.3, -602.0); (-246.1, -651.5); (-90.9, -147.6) (101.8, -579.4); (-237.3, 336.7); (-383.4, 642.7); (357.2, 276.4) (-557.3, 278.5); (87.2, 499.2); (-727.8, -150.9); (633.4, -44.0)
1	1000	15	LX-SSA-VNS	208595.1	(553.1, -788.7); (594.0, 797.5); (-373.7, -441.7); (755.7, 466.9) (314.3, -516.5); (-32.7, -887.3); (-21.2, 169.8); (555.5, -38.3) (-829.8, 384.0); (-824.4, -418.2); (172.6, 894.5); (-536.9, -761.9) (954.0, -104.0); (-549.1, 153.4); (190.2, -160.2)
2	500	10	LX-SSA-VNS	68295.1	(-214.4, 9.6); (80.6, 148.3); (-26.1, -498.9); (498.2, -41.1) (-324.5, -304.4); (59.6, -172.0); (343.2, -363.4); (186.3, 463.8) (425.3, 261.2); (-149.7, 405.8)
2	750	12	VNS	83609.6	(-220.7, -240.0); (-184.6, -704.7); (425.4, -617.0); (-196.1, 723.8) (745.5, 9.5); (502.1, -260.4); (259.7, 701.5); (40.1, 163.4) (-660.6, -195.5); (-541.5, -518.0); (-299.6, 433.0); (-722.9, 161.4)
2	1000	15	LX-SSA-VNS	105261.6	(762.4, -262.1); (-103.1, -497.2); (303.5, -952.3); (500.2, 220.1) (-859.7, -238.2); (133.9, -228.8); (-69.7, 381.7); (758.9, 611.2) (732.6, -677.0); (-213.1, 6.2); (243.5, 814.3); (-785.0, 189.5) (-759.1, 647.1); (273.2, -515.0); (-547.0, -725.4)

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